

PROJECT SEQUENCE FLOW — *BANK LOAN ANALYSIS (Power BI + SQL)*

Overview

This project involves analyzing bank loan data to provide insights into lending practices. By utilizing Power BI for visualization and SQL for data management, we aim to assess borrowing trends, risk factors, and repayment behaviors.

Key Objectives:

- Understand the factors influencing loan approval and performance.
- Visualize loan data to identify trends and insights.
- Provide a detailed analysis of borrower profiles and financial behaviors.

PHASE 1: DOMAIN UNDERSTANDING

Goal: Know *what*, *why*, and *how* the project matters in banking.

1. Introduction to Bank Loans

- Importance of bank loans for individuals & businesses.
- Types of loans (personal, business, mortgage, etc.).
- Loan lifecycle: Application → Approval → Disbursement → Repayment → Closure.

2. Why Analyze Bank Loan Data

- Risk assessment, fraud detection, portfolio health, profitability, and compliance.
- Helps in decision-making and customer segmentation.

3. Process of Granting a Loan

- Step-by-step: Application → Review → Credit Check → DTI → Approval → Disbursement → Repayment.
- Note: Link this to *data fields* you'll later analyze.

4. Sources of Loan Data

- Applications, credit bureaus, internal records, online portals, and third-party data.



DOMAIN KNOWLEDGE

Bank loans are a crucial financial tool that enable individuals and businesses to achieve their goals and manage financial needs. However, it's essential for borrowers to understand the **terms, costs, and responsibilities** associated with loans to make informed financial decisions.



How Banks Collect Loan Data

Banks collect loan data through various channels and processes, including:

1. Loan Applications

When individuals or businesses apply for loans, they submit detailed applications that include personal and financial information. This data is collected electronically or in paper form.

2. Credit Reports

Banks access credit reports from credit bureaus when assessing a borrower's creditworthiness. These reports contain information about credit history, existing loans, and payment behavior.

3. Internal Records

Banks maintain internal records of loan transactions, including **disbursements, repayments, and loan status changes**, stored in their databases.

4. Online Portals

Many banks offer online platforms where borrowers can apply for loans, make payments, and view account information. Data from these portals is collected and stored for analysis.

5. Third-party Data Sources

Banks may also use external sources, such as **income verification services**, to gather additional borrower information.



Process of Granting a Loan

1. Loan Application

The customer submits a loan application either **in person**, **online**, or through other channels.

2. Application Review

The lending institution reviews the application and collects necessary documentation such as **income statements**, **credit reports**, and **ID proofs**.

3. Identity Verification

The bank verifies the applicant's identity to prevent **identity theft** and confirm authenticity.

4. Credit Check

A credit check is performed to evaluate the applicant's **credit score**, **credit history**, and any past defaults.

5. Income Verification

Lenders verify income by reviewing **pay slips**, **tax returns**, or other documents to assess repayment ability.

6. Debt-to-Income Ratio (DTI) Check

$DTI = (\text{Monthly Debt Payments} / \text{Monthly Income})$.

A **lower DTI** indicates a better capacity to repay the loan.

7. Employment Verification

Banks may contact the applicant's employer to confirm **employment status** and **length of service**.

8. Collateral Assessment (if applicable)

If the loan is secured, lenders evaluate the **value and condition** of the collateral (e.g., home, vehicle).

9. Risk Assessment

Banks assess overall loan risk based on credit score, income, employment, and loan purpose.

10. Loan Approval or Denial

Based on the evaluation, the lender **approves or denies** the application. Approved loans are assigned **terms, interest rates, and amounts**.

11. Loan Agreement

An official **loan agreement** is created outlining:

- Interest rate
- Repayment schedule
- Fees and penalties
- Terms and conditions

12. Disbursement of Funds

Once signed, funds are disbursed to the borrower for the specified purpose.

13. Repayment

Borrowers make regular payments including **principal + interest**, as outlined in the loan agreement.

14. Ongoing Monitoring

Lenders continue to monitor:

- Payment behavior
 - Borrower's financial health
 - Delinquencies or defaults
-

Reasons for Analyzing Bank Loan Data

Banks analyze loan data for several key reasons:

1. Risk Assessment

To evaluate borrower creditworthiness, predict **default probabilities**, and determine **interest rates**.

2. Decision-making

Data-driven insights help banks **approve or deny** loans and optimize lending strategies.

3. Portfolio Management

Analysis enables tracking of portfolio performance, identifying **underperforming loans**, and adjusting terms.

4. Fraud Detection

Banks detect fraudulent activity by identifying **unusual patterns** or **data inconsistencies**.

5. Regulatory Compliance

Ensures compliance with laws like:

- **Home Mortgage Disclosure Act (HMDA)**
- **Know Your Customer (KYC)**

6. Customer Insights

Provides understanding of **customer behavior, preferences, and needs**, allowing personalized loan offers.

7. Profitability Analysis

Assesses profitability by examining:

- Interest income
- Origination costs
- Default rates
- Collection efficiency

8. Market Research

Helps understand **market trends**, **competition**, and **demand**, aiding product development and strategy.

9. Credit Risk Management

Supports **risk mitigation**, **loss provisioning**, and **stress testing** of loan portfolios.

10. Customer Retention

Identifies opportunities for:

- Loan refinancing
- Cross-selling other financial products

PHASE 2: DATA UNDERSTANDING & TERMINOLOGIES

Goal: Learn what each column means and why banks collect it.

1. Key Data Fields

- Go through **Loan ID** → **Loan Amount** → **Interest Rate** → **Loan Status**, etc.
- For each, write:
 - **Purpose (meaning)**
 - **Use for Banks (business insight)**

💡 *Tip:* Make a two-column table in your notes like

Field	Purpose / Use
Loan ID	Unique loan identifier
DTI	Measures borrower's financial burden

2. Important Metrics

- Total Loan Applications
- Funded Amount
- Amount Received
- Average Interest Rate
- Average DTI

These will later become **KPIs** in Power BI.



Terminologies Used in Data

1. Loan ID

- **Purpose:** Unique identifier assigned to each loan application or account; acts as a primary key for tracking and management.
- **Use for Banks:** Helps efficiently manage, track, and organize loan records throughout their lifecycle.

2. Address State

- **Purpose:** Indicates the borrower's location; helps assess regional risk factors and compliance with regulations.
- **Use for Banks:** Identifies regional loan trends, adjusts marketing strategies, and manages geographic risk portfolios.

3. Employee Length

- **Purpose:** Reflects the borrower's employment stability; longer tenure indicates greater job security.

- **Use for Banks:** Assists in assessing repayment ability — longer employment often means lower default risk.
-

4. Employee Title

- **Purpose:** Specifies the borrower's job title or occupation; helps identify income sources.
 - **Use for Banks:** Verifies income and tailors loan products for different professions.
-

5. Grade

- **Purpose:** Risk classification assigned to a loan based on creditworthiness; higher grade = lower risk.
 - **Use for Banks:** Used to price loans and manage portfolio risk; higher grades get lower interest rates.
-

6. Sub Grade

- **Purpose:** Provides further granularity within a grade for precise risk differentiation.
 - **Use for Banks:** Enables fine-tuned interest rates and terms based on borrower risk profiles.
-

7. Home Ownership

- **Purpose:** Indicates the borrower's housing status, showing financial stability.
 - **Use for Banks:** Assesses collateral potential and borrower stability; homeowners often have lower default rates.
-

8. Issue Date

- **Purpose:** Marks the loan origination date; crucial for tracking and maturity calculations.

- **Use for Banks:** Helps track loan aging, interest accruals, and portfolio management.
-

9. Last Credit Pull Date

- **Purpose:** Records when the borrower's credit was last checked.
 - **Use for Banks:** Monitors updates in creditworthiness and supports credit risk assessments.
-

10. Last Payment Date

- **Purpose:** Indicates the most recent loan payment date.
 - **Use for Banks:** Tracks borrower payment behavior, identifies delinquencies, and forecasts payments.
-

11. Loan Status

- **Purpose:** Describes the current loan state (e.g., *Fully Paid*, *Current*, *Default*).
 - **Use for Banks:** Used to monitor loan performance and categorize loans for risk analysis.
-

12. Next Payment Date

- **Purpose:** Predicts the next scheduled payment date.
 - **Use for Banks:** Aids in cash flow forecasting and revenue planning.
-

13. Purpose

- **Purpose:** Defines the reason for the loan (e.g., *Debt Consolidation*, *Education*).
 - **Use for Banks:** Helps segment borrowers and customize loan offerings.
-

14. Term

- **Purpose:** Specifies the duration of the loan (in months).

- **Use for Banks:** Determines repayment period, interest calculation, and maturity management.
-

15. Verification Status

- **Purpose:** Shows if the borrower's information has been verified.
 - **Use for Banks:** Measures data credibility and supports accurate loan evaluations.
-

16. Annual Income

- **Purpose:** Displays total yearly earnings of the borrower.
 - **Use for Banks:** Used to assess loan eligibility and calculate debt-to-income ratio.
-

17. DTI (Debt-to-Income Ratio)

- **Purpose:** Ratio showing debt burden relative to income.
 - **Use for Banks:** Evaluates borrower's capacity to handle more debt responsibly.
-

18. Installment

- **Purpose:** Fixed monthly payment amount (principal + interest).
 - **Use for Banks:** Determines payment schedules and affordability.
-

19. Interest Rate

- **Purpose:** Annual borrowing cost expressed as a percentage.
 - **Use for Banks:** Used to set loan pricing, manage profits, and attract investors.
-

20. Loan Amount

- **Purpose:** Total amount borrowed (principal).

- **Use for Banks:** Defines the loan size and helps in capital allocation and portfolio planning.



PHASE 3: DATA HANDLING IN SQL



SQL Workflow (Data Extraction)

This phase involves handling and extracting bank loan data using **Microsoft SQL Server Management Studio (SSMS)**.

SQL – MS SQL SERVER

- ✓ Creating Database
- ✓ Creating Table
- ✓ Select
- ✓ Datename
- ✓ Datepart
- ✓ Cast
- ✓ Decimal
- ✓ Month
- ✓ Hour
- ✓ Quarter
- ✓ Day
- ✓ Group by
- ✓ Order by
- ✓ Decimal
- ✓ Limit
- ✓ Count
- ✓ Distinct
- ✓ CTE
- ✓ Partition

Step-by-Step Workflow

1. Open SQL Server Management Studio (SSMS)

Launch SSMS and connect to your SQL Server instance.

2. Create a New Database

Create a database named `BANK_LOAN` using the following command:

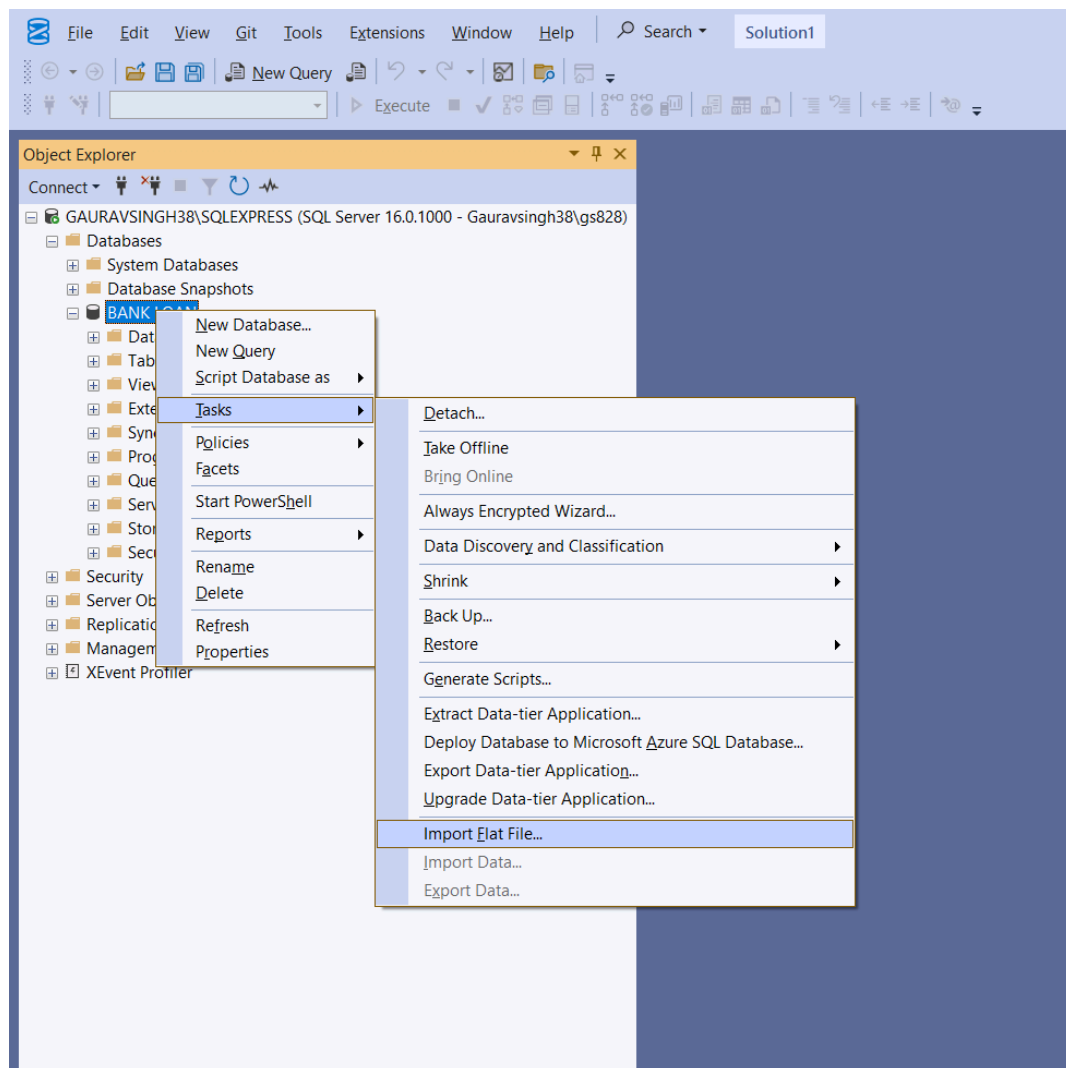

```
CREATE DATABASE BANK_LOAN;
```

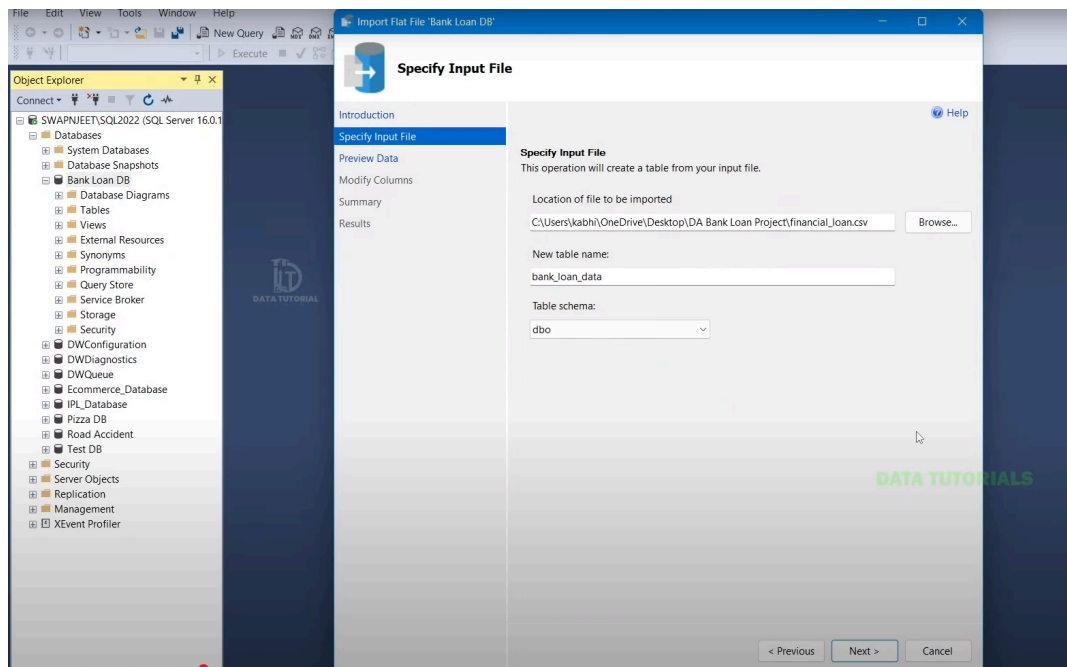
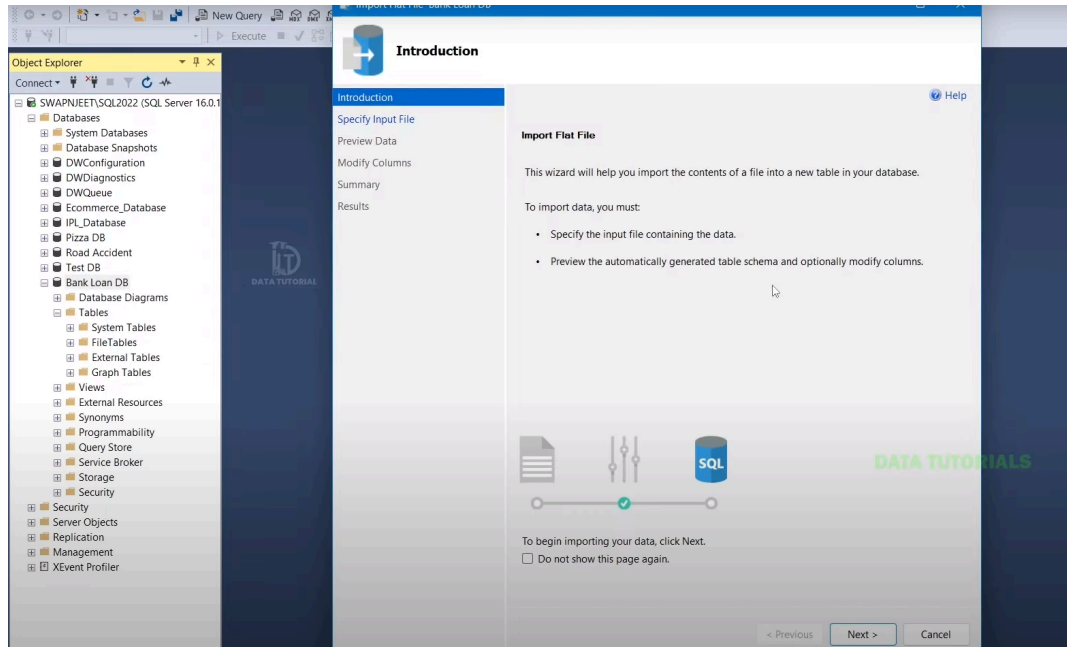
3. Create Table (Structure as per Provided Schema/Photos)

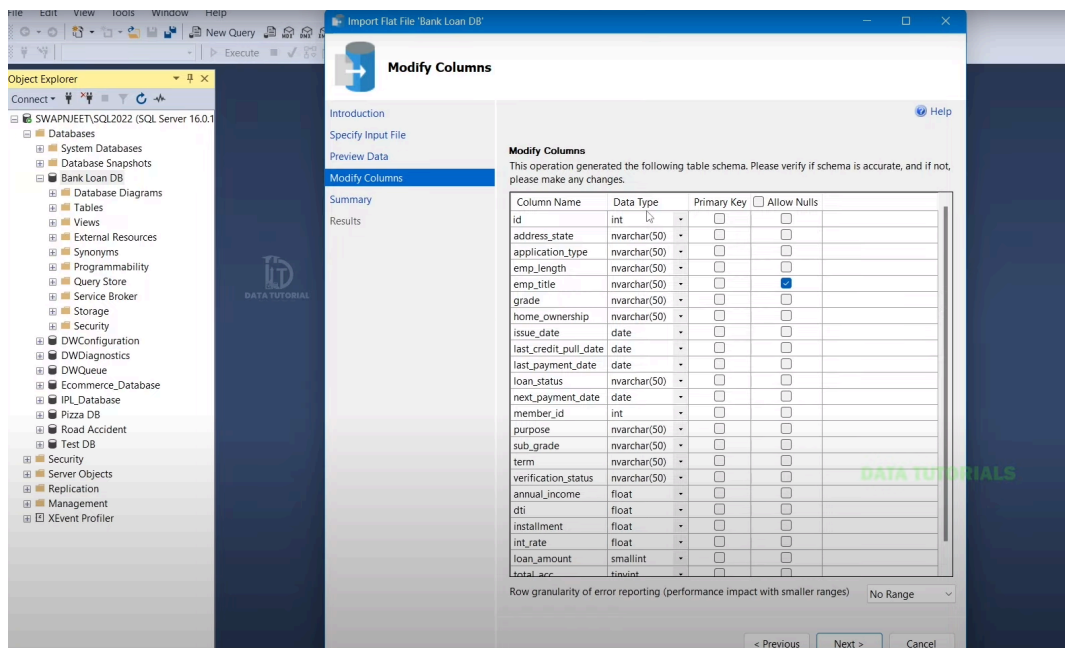
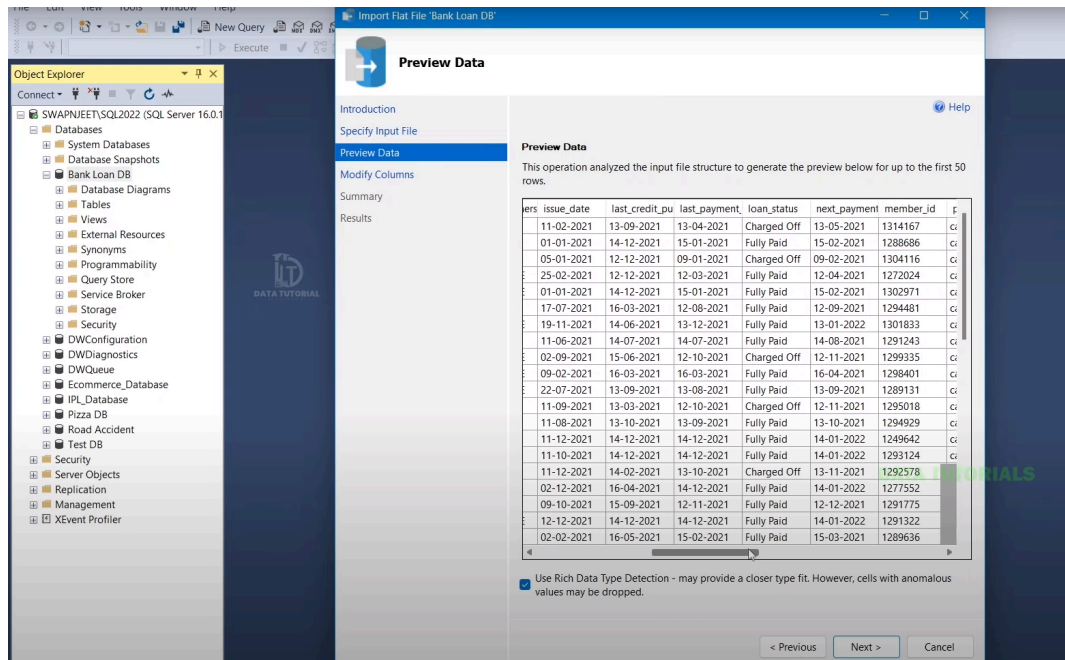
In the **BANK_LOAN** database, create a table to store loan data.

(Use the table structure given in your dataset or photos.)

Example:







4. Import Data into SQL Table

- Import your dataset (e.g., CSV or Excel) into the `bank_loan_data` table.
- During import, verify:
 - ☒ Correct **data types** (e.g., numeric, date, varchar)
 - ☒ Proper **constraints** (e.g., primary key, not null)

Columns	
id	(PK, int, not null)
address_state	(varchar(50), not null)
application_type	(varchar(50), not null)
emp_length	(varchar(50), not null)
emp_title	(varchar(100), null)
grade	(varchar(50), not null)
home_ownership	(varchar(50), not null)
issue_date	(date, not null)
last_credit_pull_date	(date, not null)
last_payment_date	(date, not null)
loan_status	(nvarchar(50), not null)
next_payment_date	(date, not null)
member_id	(int, not null)
purpose	(varchar(50), not null)
sub_grade	(varchar(50), not null)
term	(varchar(50), not null)
verification_status	(varchar(50), not null)
annual_income	(float, not null)
dti	(float, not null)
installment	(float, not null)
int_rate	(float, not null)
loan_amount	(int, not null)
total_acc	(tinyint, not null)
total_payment	(int, not null)

5. Verify Imported Data

Run a simple query to check if your data has been imported correctly:

```
SELECT * FROM bank_loan_data;
```

6. Validate Record Count

To ensure complete import, check total rows:

```
SELECT COUNT(*) AS Total_Records FROM bank_loan_data;
```

7. Check Column Metadata (Optional)

To verify column names, data types, and constraints:

```
EXEC sp_help 'bank_loan_data';
```

Business Problem Solving

PROBLEM STATEMENT

DASHBOARD 1: SUMMARY

Key Performance Indicators (KPIs) Requirements:

- Total Loan Applications:** We need to calculate the total number of loan applications received during a specified period. Additionally, it is essential to monitor the Month-to-Date (MTD) Loan Applications and track changes Month-over-Month (MoM).
- Total Funded Amount:** Understanding the total amount of funds disbursed as loans is crucial. We also want to keep an eye on the MTD Total Funded Amount and analyse the Month-over-Month (MoM) changes in this metric.
- Total Amount Received:** Tracking the total amount received from borrowers is essential for assessing the bank's cash flow and loan repayment. We should analyse the Month-to-Date (MTD) Total Amount Received and observe the Month-over-Month (MoM) changes.
- Average Interest Rate:** Calculating the average interest rate across all loans, MTD, and monitoring the Month-over-Month (MoM) variations in interest rates will provide insights into our lending portfolio's overall cost.
- Average Debt-to-Income Ratio (DTI):** Evaluating the average DTI for our borrowers helps us gauge their financial health. We need to compute the average DTI for all loans, MTD, and track Month-over-Month (MoM) fluctuations.



Understanding MTD ,PMTD and MoM



1. Month-to-Date (MTD)

Meaning:

Month-to-Date (MTD) refers to the period starting from the **first day of the current month up to today's date**.

It measures performance, growth, or totals within the **ongoing month**.

Example:

If today is **October 5**,
then **MTD** covers the period from **October 1 → October 5**.

Use Case:

Banks or analysts use MTD to track:

- 🏦 Total loan disbursements so far this month
 - 💰 Revenue or repayments up to today
 - 📈 KPI progress (like active loans, defaults, etc.)
-



2. PMTD – Previous Month-to-Date



Meaning

PMTD (Previous Month-to-Date) refers to the same number of days in the **previous month** as the current date in the **current month**.

It allows comparison between the current month's performance (MTD) and the same period from the **previous month**.



Example:

If **today is October 5**,
then:

- **MTD period:** October 1 → October 5
- **PMTD period:** September 1 → September 5

✅ This helps you compare *how your KPIs performed in the same duration last month*.



Use Case

Banks and analysts use **PMTD** to:

- Compare **month-to-date** performance with the same time in the previous month

- Identify early trends (growth or decline)
- Support **MoM analysis** before the current month ends

For example:

If total disbursed loans (MTD) = ₹8,00,000
 and previous month-to-date loans (PMTD) = ₹7,50,000
 → MTD vs PMTD Growth = **6.67% increase**

3. Month-over-Month (MoM)

Meaning:

Month-over-Month (MoM) compares performance between **two consecutive months** — typically used to analyze **growth or decline trends**.

It helps understand **how a metric changed** from one month to the next.

Formula:

$$\text{MoM Growth (\%)} = \frac{\text{Current Month Value} - \text{Previous Month Value}}{\text{Previous Month Value}} \times 100$$

Example:

- September Loan Disbursed = ₹10,00,000
- October Loan Disbursed = ₹12,00,000

$$\text{MoM Growth} = \frac{(12,00,000 - 10,00,000)}{10,00,000} \times 100 = 20\% \quad \text{\text{MoM Growth} = \frac{(12,00,000 - 10,00,000)}{10,00,000} \times 100 = 20\%}$$

$$\text{MoM Growth} = \frac{10,00,000(12,00,000 - 10,00,000)}{10,00,000} \times 100 = 20\%$$

So, there is a **20% increase** in disbursed loans compared to last month.

Use Case:

-  Track monthly loan growth or repayments
-  Identify rising default rates

- 📊 Monitor revenue or new customer acquisition month by month

BANK LOAN REPORT QUERY DOCUMENT

A. BANK LOAN REPORT | SUMMARY

KPI's:

Total Loan Applications

```
SELECT COUNT(id) AS Total_Applications FROM bank_loan_data
```

Total_Applications
38576

MTD Loan Applications

```
SELECT COUNT(id) AS MTD_Total_Applications FROM bank_loan_data  
WHERE MONTH(issue_date) = 12
```

Total_Applications
4314

PMTD Loan Applications

```
SELECT COUNT(id) AS PMTD_Total_Applications FROM bank_loan_data  
WHERE MONTH(issue_date) = 11 AND YEAR(issue_date)=2021
```

Total_Applications
4035

Total Funded Amount

```
SELECT SUM(loan_amount) AS Total_Funded_Amount FROM bank_loan_data
```

Total_Funded_Amount
435757075

MTD Total Funded Amount

```
SELECT SUM(loan_amount) AS MTD_Total_Funded_Amount FROM bank_loan_data  
WHERE MONTH(issue_date) = 12
```

Total_Funded_Amount
53981425

PMTD Total Funded Amount

```
SELECT SUM(loan_amount) AS PMTD_Total_Funded_Amount FROM bank_loan_data  
WHERE MONTH(issue_date) = 11
```

Total_Funded_Amount
47754825

Total Amount Received

```
SELECT SUM(total_payment) AS Total_Amount_Collected FROM bank_loan_data
```

Total_Amount_Collected
473070933

MTD Total Amount Received

```
SELECT SUM(total_payment) AS MTD_Total_Amount_Collected FROM  
bank_loan_data
```

```
WHERE MONTH(issue_date) = 12
```

Total_Amount_Collected
58074380

PMTD Total Amount Received

```
SELECT SUM(total_payment) AS PMTD_Total_Amount_Collected FROM
bank_loan_data
WHERE MONTH(issue_date) = 11
```

Total_Amount_Collected
50132030

Average Interest Rate

```
SELECT ROUND(AVG(int_rate)*100,4) AS Avg_Int_Rate FROM bank_loan_data
```

Avg_Int_Rate
12.0488314172048

MTD Average Interest

```
SELECT ROUND(AVG(int_rate)*100,4) AS MTD_Avg_Int_Rate FROM bank_loan_data
WHERE MONTH(issue_date) = 12
```

MTD_Avg_Int_Rate
12.3560408676042

PMTD Average Interest

```
SELECT ROUND(AVG(int_rate)*100,4) AS PMTD_Avg_Int_Rate FROM bank_loan_data
WHERE MONTH(issue_date) = 11
```

PMTD_Avg_Int_Rate
11.9417175498261

Avg DTI

```
SELECT ROUND(AVG(dti)*100,4) AS Avg_DTI FROM bank_loan_data
```

Avg_DTI
13.3274331211432

MTD Avg DTI

```
SELECT ROUND(AVG(dti)*100,4) AS MTD_Avg_DTI FROM bank_loan_data  
WHERE MONTH(issue_date) = 12
```

MTD_Avg_DTI
13.6655377880425

PMTD Avg DTI

```
SELECT ROUND(AVG(dti)*100,4) AS PMTD_Avg_DTI FROM bank_loan_data  
WHERE MONTH(issue_date) = 11
```

PMTD_Avg_DTI
13.3027335836364

PROBLEM STATEMENT



DASHBOARD 1: SUMMARY

Good Loan v Bad Loan KPI's

Good Loan:

1. Good Loan Application Percentage
2. Good Loan Applications
3. Good Loan Funded Amount
4. Good Loan Total Received Amount

Bad Loan

1. Bad Loan Application Percentage
2. Bad Loan Applications
3. Bad Loan Funded Amount
4. Bad Loan Total Received Amount

Loan Status Grid View

In order to gain a comprehensive overview of our lending operations and monitor the performance of loans, we aim to create a grid view report categorized by 'Loan Status.' By providing insights into metrics such as 'Total Loan Applications,' 'Total Funded Amount,' 'Total Amount Received,' 'Month-to-Date (MTD) Funded Amount,' 'MTD Amount Received,' 'Average Interest Rate,' and 'Average Debt-to-Income Ratio (DTI),' this grid view will empower us to make data-driven decisions and assess the health of our loan portfolio.

GOOD LOAN ISSUED

Good Loan Percentage

```
SELECT  
    (COUNT(CASE WHEN loan_status = 'Fully Paid' OR loan_status = 'Current' THEN id  
END) * 100.0) /  
    COUNT(id) AS Good_Loan_Percentage  
FROM bank_loan_data
```

Good_Loan_Percentage
86.175342181667

Good Loan Applications

```
SELECT COUNT(id) AS Good_Loan_Applications FROM bank_loan_data  
WHERE loan_status = 'Fully Paid' OR loan_status = 'Current'
```

Good_Loan_Applications
33243

Good Loan Funded Amount

```
SELECT SUM(loan_amount) AS Good_Loan_Funded_amount FROM bank_loan_data  
WHERE loan_status = 'Fully Paid' OR loan_status = 'Current'
```

Good_Loan_Funded_amount
370224850

Good Loan Amount Received

```
SELECT SUM(total_payment) AS Good_Loan_amount_received FROM bank_loan_data  
WHERE loan_status = 'Fully Paid' OR loan_status = 'Current'
```

Good_Loan_amount_received
435786170

BAD LOAN ISSUED

Bad Loan Percentage

```
SELECT  
    (COUNT(CASE WHEN loan_status = 'Charged Off' THEN id END) * 100.0) /  
    COUNT(id) AS Bad_Loan_Percentage  
FROM bank_loan_data
```

Bad_Loan_Percentage
13.824657818332

Bad Loan Applications

```
SELECT COUNT(id) AS Bad_Loan_Applications FROM bank_loan_data  
WHERE loan_status = 'Charged Off'
```

Bad_Loan_Applications
5333

Bad Loan Funded Amount

```
SELECT SUM(loan_amount) AS Bad_Loan_Funded_amount FROM bank_loan_data  
WHERE loan_status = 'Charged Off'
```

Bad_Loan_Funded_amount
65532225

Bad Loan Amount Received

```
SELECT SUM(total_payment) AS Bad_Loan_amount_received FROM bank_loan_data  
WHERE loan_status = 'Charged Off'
```

Bad_Loan_amount_received
37284763

LOAN STATUS

```

SELECT
    loan_status,
    COUNT(id) AS LoanCount,
    SUM(total_payment) AS Total_Amount_Received,
    SUM(loan_amount) AS Total_Funded_Amount,
    AVG(int_rate * 100) AS Interest_Rate,
    AVG(dti * 100) AS DTI
FROM
    bank_loan_data
GROUP BY
    loan_status

```

	loan_status	LoanCount	Total_Amount_Received	Total_Funded_Amount	Interest_Rate	DTI
1	Fully Paid	32145	411586256	351358350	11.6410707918092	13.1673507557434
2	Charged Off	5333	37284763	65532225	13.8785749318289	14.0047328005517
3	Current	1098	24199914	18866500	15.0993260800947	14.7243442736843

```

SELECT
    loan_status,
    SUM(total_payment) AS MTD_Total_Amount_Received,
    SUM(loan_amount) AS MTD_Total_Funded_Amount
FROM bank_loan_data
WHERE MONTH(issue_date) = 12
GROUP BY loan_status

```

loan_status	MTD_Total_Amount_Received	MTD_Total_Funded_Amount
Fully Paid	47815851	41302025
Charged Off	5324211	8732775
Current	4934318	3946625

PROBLEM STATEMENT



DASHBOARD 2: OVERVIEW

CHARTS

1. **Monthly Trends by Issue Date (Line Chart):** To identify seasonality and long-term trends in lending activities
2. **Regional Analysis by State (Filled Map):** To identify regions with significant lending activity and assess regional disparities
3. **Loan Term Analysis (Donut Chart):** To allow the client to understand the distribution of loans across various term lengths.
4. **Employee Length Analysis (Bar Chart):** How lending metrics are distributed among borrowers with different employment lengths, helping us assess the impact of employment history on loan applications.
5. **Loan Purpose Breakdown (Bar Chart):** Will provide a visual breakdown of loan metrics based on the stated purposes of loans, aiding in the understanding of the primary reasons borrowers seek financing.
6. **Home Ownership Analysis (Tree Map):** For a hierarchical view of how home ownership impacts loan applications and disbursements.

Metrics to be shown: 'Total Loan Applications,' 'Total Funded Amount,' and 'Total Amount Received'

B. BANK LOAN REPORT | OVERVIEW

MONTH

```
SELECT
    MONTH(issue_date) AS Month_Munber,
    DATENAME(MONTH, issue_date) AS Month_name,
    COUNT(id) AS Total_Loan_Applications,
    SUM(loan_amount) AS Total_Funded_Amount,
    SUM(total_payment) AS Total_Amount_Received
FROM bank_loan_data
GROUP BY MONTH(issue_date), DATENAME(MONTH, issue_date)
ORDER BY MONTH(issue_date)
```

	Month_Munber	Month_name	Total_Loan_Applications	Total_Funded_Amount	Total_Amount_Received
1	1	January	2332	25031650	27578836
2	2	February	2279	24647825	27717745
3	3	March	2627	28875700	32264400
4	4	April	2755	29800800	32495533
5	5	May	2911	31738350	33750523
6	6	June	3184	34161475	36164533
7	7	July	3366	35813900	38827220
8	8	August	3441	38149600	42682218
9	9	September	3536	40907725	43983948
10	10	October	3796	44893800	49399567
11	11	November	4035	47754825	50132030
12	12	December	4314	53981425	58074380

STATE

```
SELECT
    address_state AS State,
    COUNT(id) AS Total_Loan_Applications,
    SUM(loan_amount) AS Total_Funded_Amount,
    SUM(total_payment) AS Total_Amount_Received
FROM bank_loan_data
GROUP BY address_state
ORDER BY address_state
```

	State	Total_Loan_Applications	Total_Funded_Amount	Total_Amount_Received
1	AK	78	1031800	1108570
2	AL	432	4949225	5492272
3	AR	236	2529700	2777875
4	AZ	833	9206000	10041986
5	CA	6894	78484125	83901234
6	CO	770	8976000	9845810
7	CT	730	8435575	9357612
8	DC	214	2652350	2921854
9	DE	110	1138100	1269136
10	FL	2773	30046125	31601905
11	GA	1355	15480325	16728040
12	HI	170	1850525	2080184
13	IA	5	56450	64482
14	ID	6	59750	65329
15	IL	1486	17124225	18875941
16	IN	9	86225	85521
17	KS	260	2872325	3247394
18	KY	320	3504100	3792530

TERM

```
SELECT
    term AS Term,
    COUNT(id) AS Total_Loan_Applications,
    SUM(loan_amount) AS Total_Funded_Amount,
    SUM(total_payment) AS Total_Amount_Received
FROM bank_loan_data
GROUP BY term
ORDER BY term
```

	Term	Total_Loan_Applications	Total_Funded_Amount	Total_Amount_Received
1	36 months	28237	273041225	294709458
2	60 months	10339	162715850	178361475

EMPLOYEE LENGTH

```
SELECT
    emp_length AS Employee_Length,
    COUNT(id) AS Total_Loan_Applications,
    SUM(loan_amount) AS Total_Funded_Amount,
    SUM(total_payment) AS Total_Amount_Received
FROM bank_loan_data
GROUP BY emp_length
ORDER BY emp_length
```

Employee_Length	Total_Loan_Applications	Total_Funded_Amount	Total_Amount_Received
< 1 year	4575	44210625	47545011
1 year	3229	32883125	35498348
10+ years	8870	116115950	125871616
2 years	4382	44967975	49206961
3 years	4088	43937850	47551832
4 years	3428	37600375	40964850
5 years	3273	36973625	40397571
6 years	2228	25612650	27908658
7 years	1772	20811725	22584136
8 years	1476	17558950	19025777
9 years	1255	15084225	16516173

PURPOSE

```
SELECT
    purpose AS PURPOSE,
    COUNT(id) AS Total_Loan_Applications,
    SUM(loan_amount) AS Total_Funded_Amount,
    SUM(total_payment) AS Total_Amount_Received
FROM bank_loan_data
GROUP BY purpose
ORDER BY purpose
```

PURPOSE	Total_Loan_Applications	Total_Funded_Amount	Total_Amount_Received
car	1497	10223575	11324914
credit card	4998	58885175	65214084
Debt consolidation	18214	232459675	253801871
educational	315	2161650	2248380
home improvement	2876	33350775	36380930
house	366	4824925	5185538
major purchase	2110	17251600	18676927
medical	667	5533225	5851372
moving	559	3748125	3999899
other	3824	31155750	33289676
renewable_energy	94	845750	898931
small business	1776	24123100	23814817
vacation	352	1967950	2116738
wedding	928	9225800	10266856

HOME OWNERSHIP

```

SELECT
    home_ownership AS Home_Ownership,
    COUNT(id) AS Total_Loan_Applications,
    SUM(loan_amount) AS Total_Funded_Amount,
    SUM(total_payment) AS Total_Amount_Received
FROM bank_loan_data
GROUP BY home_ownership
ORDER BY home_ownership

```

Home_Ownership	Total_Loan_Applications	Total_Funded_Amount	Total_Amount_Received
MORTGAGE	17198	219329150	238474438
NONE	3	16800	19053
OTHER	98	1044975	1025257
OWN	2838	29597675	31729129
RENT	18439	185768475	201823056

PROBLEM STATEMENT

DASHBOARD 3: DETAILS



GRID

Need for a comprehensive 'Details Dashboard' that provides a consolidated view of all the essential information within our loan data. This Details Dashboard aims to offer a holistic snapshot of key loan-related metrics and data points, enabling users to access critical information efficiently.

Objective:

The primary objective of the Details Dashboard is to provide a comprehensive and user-friendly interface for accessing vital loan data. It will serve as a one-stop solution for users seeking detailed insights into our loan portfolio, borrower profiles, and loan performance.

C. BANK LOAN REPORT | DETAILS

SQLQuery1.s... Executing...*																
1 select * from bank_loan_data																
100 % No issues found																
	id	address_state	application_type	emp_length	emp_title	grade	home_ownership	issue_date	last_credt_lpuil_date	last_payment_date	loan_status	next_payment_date	member_id	purpose	sub_grade	term
1	54734	CA	INDIVIDUAL	< 1 year	NULL	B	RENT	2021-08-09	2021-08-12	2021-10-11	Fully Paid	2021-11-11	80364	Debt consolidation	B4	36 months
2	55742	NY	INDIVIDUAL	< 1 year	CNN	B	RENT	2021-05-08	2021-05-12	2021-05-11	Fully Paid	2021-07-11	114426	credit card	B5	36 months
3	57245	TX	INDIVIDUAL	10+ years	city of Beaumont Texas	C	OWN	2021-03-10	2021-05-16	2021-03-13	Fully Paid	2021-04-13	138150	Debt consolidation	C2	36 months
4	57416	CT	INDIVIDUAL	6 years	State Farm Insurance	C	RENT	2021-11-09	2021-05-16	2021-11-12	Fully Paid	2021-12-12	139635	Debt consolidation	C3	36 months
5	58915	CA	INDIVIDUAL	3 years	GLJalcom Inc	B	RENT	2021-04-08	2021-03-14	2021-04-11	Fully Paid	2021-05-11	153417	Debt consolidation	B3	36 months
6	59006	TX	INDIVIDUAL	3 years	NULL	C	MORTGAGE	2021-09-09	2021-09-12	2021-10-12	Fully Paid	2021-11-12	154254	credit card	C5	36 months
7	61390	TX	INDIVIDUAL	< 1 year	NULL	A	MORTGAGE	2021-02-10	2021-03-12	2021-03-12	Fully Paid	2021-04-12	162594	credit card	A5	36 months
8	61419	MD	INDIVIDUAL	1 year	Pension Benefit Guaranty Corporation	D	RENT	2021-02-10	2021-03-13	2021-10-12	Charged Off	2021-11-12	162917	Debt consolidation	D2	36 months
9	62102	MA	INDIVIDUAL	5 years	Rockwell Automation Inc.	B	RENT	2021-04-10	2021-03-13	2021-02-11	Fully Paid	2021-03-11	191024	Debt consolidation	B1	36 months
10	65426	MI	INDIVIDUAL	< 1 year	Infotrieve, Inc.	B	MORTGAGE	2021-08-09	2021-05-16	2021-06-11	Charged Off	2021-07-11	232106	car	B1	36 months
11	65640	CA	INDIVIDUAL	10+ years	Imixionvision	C	MORTGAGE	2021-06-08	2021-04-15	2021-06-11	Fully Paid	2021-06-11	234969	home improvement	C2	36 months
12	66431	KY	INDIVIDUAL	3 years	Video Monitoring Services	B	RENT	2021-02-08	2021-03-12	2021-03-12	Fully Paid	2021-04-12	243540	Debt consolidation	B5	36 months
13	66749	MS	INDIVIDUAL	10+ years	crown cork & seal	C	MORTGAGE	2021-12-08	2021-12-11	2021-01-12	Fully Paid	2021-02-12	246329	Debt consolidation	C4	36 months
14	66943	CA	INDIVIDUAL	< 1 year	The Coggins Co.	B	RENT	2021-08-10	2021-07-14	2021-10-11	Fully Paid	2021-11-11	247802	Debt consolidation	B4	60 months
15	66964	KS	INDIVIDUAL	3 years	Acquity	D	MORTGAGE	2021-06-08	2021-06-11	2021-06-11	Fully Paid	2021-07-11	247990	Debt consolidation	D3	36 months
16	67503	MI	INDIVIDUAL	1 year	Genesis PHD	A	MORTGAGE	2021-10-09	2021-02-14	2021-02-12	Fully Paid	2021-03-12	252415	Debt consolidation	A4	36 months

PHASE 4: DATA VISUALIZATION IN POWER BI

If you are facing issue while connecting to MS SQL Server then you connect to csv file as well and complete the project. Just click on Get Data > Text/CSV > Locate the File you Downloaded. You will same results as in video by using text file connection.

Goal: Build dynamic, business-ready dashboards.

POWER BI

- ✓ Connecting to SQL Server
- ✓ Data Cleaning
- ✓ Data Modelling
- ✓ Data Processing
- ✓ Power Query
- ✓ Date Tables
- ✓ Time Intelligence Func
- ✓ DAX
- ✓ Date Function
- ✓ Text Function
- ✓ Filter Function
- ✓ Calculate
- ✓ SUM/ SUMX
- ✓ Creating KPI's
- ✓ New Card Visual
- ✓ Creating Charts
- ✓ Formatting visuals
- ✓ Creating Functions
- ✓ Navigations

DATA QUALITY CHECK & CLEANING (Power Query Editor)

1 Open Power Query Editor

- In Power BI → **Home Tab** → **Transform Data** → opens **Power Query Editor**.
- This is where you clean and prepare data before visualization.

2 Enable Data Profiling Tools

Go to the **View tab** → tick options:

-  **Column Quality** → shows % of valid, error, and empty data.
-  **Column Distribution** → shows unique & distinct values (top 1000 rows by default).
-  **Column Profile** (optional) → full statistical summary for selected columns.

 *Tip:* Check for duplicates or missing data in your **Loan_ID** column — your **primary key**.

You can also check for **white spaces** or **data type mismatches** (e.g., numbers stored as text).

DATA CLEANING

- Remove nulls or blanks in crucial columns (Loan_ID, Loan_Status, Issue_Date, etc.).
- Convert columns to proper **data types** (Date, Whole Number, Decimal, Text).
- Rename columns for readability (e.g., `int_rate` → `Interest Rate`).
- Trim whitespace: *Transform tab* → *Format* → *Trim / Clean*.

id	address_state	application_type	emp_length	emp_title	grade	home_ownership	issue_date	last_credit_pull_date	last_payment_date	loan_status
331459	CA	INDIVIDUAL	7 years	Jabil Circuit	B	RENT	08 April 2021	16 May 2021	11 April 2021	Fully Paid
400097	CA	INDIVIDUAL	4 years	PMZ	B	RENT	09 May 2021	16 May 2021	12 May 2021	Fully Paid
401782	CA	INDIVIDUAL	2 years	Tri-City Technologies	B	RENT	09 July 2021	16 May 2021	12 July 2021	Fully Paid
416011	CA	INDIVIDUAL	3 years	Placer County Water Agency	B	RENT	09 June 2021	16 May 2021	12 July 2021	Fully Paid
417906	CA	INDIVIDUAL	5 years	Merced City School District	B	RENT	10 May 2021	16 May 2021	12 March 2021	Fully Paid
433880	CA	INDIVIDUAL	6 years	LA County Probation Dept.	B	RENT	09 August 2021	16 May 2021	11 October 2021	Fully Paid
441457	CA	INDIVIDUAL	2 years	International Checkout	B	RENT	09 September 2021	16 May 2021	12 May 2021	Fully Paid
449337	CA	INDIVIDUAL	9 years	Socket Mobile, Inc.	B	RENT	09 October 2021	16 May 2021	10 October 2021	Fully Paid
452391	CA	INDIVIDUAL	4 years	Siemens	B	RENT	09 November 2021	16 May 2021	11 July 2021	Fully Paid
453894	CA	INDIVIDUAL	4 years	Sprint	B	RENT	09 November 2021	16 May 2021	12 September 2021	Fully Paid
463417	CA	INDIVIDUAL	8 years	INTERNATIONAL FREIGHT SERVICES	B	RENT	10 February 2021	16 May 2021	13 January 2021	Fully Paid
475255	CA	INDIVIDUAL	5 years	datastream	B	RENT	10 January 2021	16 May 2021	10 November 2021	Fully Paid
481679	CA	INDIVIDUAL	5 years	Rowen Holdings, Inc. (dba gSEMI)	B	RENT	10 February 2021	16 May 2021	11 August 2021	Fully Paid
483320	CA	INDIVIDUAL	9 years	Kaiser Permanente	B	RENT	10 February 2021	16 May 2021	12 July 2021	Fully Paid
490290	CA	INDIVIDUAL	7 years	Carefusion	B	RENT	10 March 2021	16 May 2021	13 March 2021	Fully Paid
497147	CA	INDIVIDUAL	2 years	Gospir Insurance Services	B	RENT	10 April 2021	16 May 2021	13 March 2021	Fully Paid
500008	CA	INDIVIDUAL	8 years	State Compensation Insurance Fund	B	RENT	10 April 2021	16 May 2021	13 April 2021	Fully Paid
507410	CA	INDIVIDUAL	2 years	Ca. Dept. of Corrections	B	RENT	10 May 2021	16 May 2021	13 June 2021	Fully Paid
513262	CA	INDIVIDUAL	1 year	Analysis Group, Inc.	B	RENT	10 May 2021	16 May 2021	13 May 2021	Fully Paid
513889	CA	INDIVIDUAL	10+ years	city of san francisco	B	RENT	10 May 2021	16 May 2021	13 May 2021	Fully Paid
525162	CA	INDIVIDUAL	3 years	Invesco	B	RENT	10 June 2021	16 May 2021	12 February 2021	Fully Paid
530154	CA	INDIVIDUAL	1 year	Esilor of America	B	RENT	10 June 2021	16 May 2021	13 July 2021	Fully Paid
540740	CA	INDIVIDUAL	2 years	Bare Back Grill Gaslamp	B	RENT	10 July 2021	16 May 2021	13 February 2021	Fully Paid
553185	CA	INDIVIDUAL	2 years	Internet Brands	B	RENT	10 July 2021	16 May 2021	12 June 2021	Fully Paid
554119	CA	INDIVIDUAL	3 years	Los Angeles Opera Company	B	RENT	10 July 2021	16 May 2021	13 August 2021	Fully Paid
560674	CA	INDIVIDUAL	5 years	LA County agricultural commissioner	B	RENT	10 August 2021	16 May 2021	13 May 2021	Fully Paid
588575	CA	INDIVIDUAL	1 year	Starrford School	B	RENT	10 September 2021	16 May 2021	13 October 2021	Fully Paid
593147	CA	INDIVIDUAL	8 years	Mid-Coast Fire Protection, Inc.	B	RENT	10 October 2021	16 May 2021	12 July 2021	Fully Paid
615460	CA	INDIVIDUAL	0 years	Convention-Carmichael	B	RENT	10 November 2021	16 May 2021	12 August 2021	Fully Paid

id	address_state	application_type	emp_length	emp_title	grade
1000	Valid	Valid	Valid	Valid	Valid
1001	Valid	Valid	Valid	Valid	Valid
1002	Valid	Valid	Valid	Valid	Valid
1003	Valid	Valid	Valid	Valid	Valid
1004	Valid	Valid	Valid	Valid	Valid
1005	Valid	Valid	Valid	Valid	Valid
1006	Valid	Valid	Valid	Valid	Valid
1007	Valid	Valid	Valid	Valid	Valid
1008	Valid	Valid	Valid	Valid	Valid
1009	Valid	Valid	Valid	Valid	Valid
1010	Valid	Valid	Valid	Valid	Valid
1011	Valid	Valid	Valid	Valid	Valid
1012	Valid	Valid	Valid	Valid	Valid
1013	Valid	Valid	Valid	Valid	Valid
1014	Valid	Valid	Valid	Valid	Valid
1015	Valid	Valid	Valid	Valid	Valid
1016	Valid	Valid	Valid	Valid	Valid
1017	Valid	Valid	Valid	Valid	Valid
1018	Valid	Valid	Valid	Valid	Valid
1019	Valid	Valid	Valid	Valid	Valid
1020	Valid	Valid	Valid	Valid	Valid
1021	Valid	Valid	Valid	Valid	Valid
1022	Valid	Valid	Valid	Valid	Valid
1023	Valid	Valid	Valid	Valid	Valid
1024	Valid	Valid	Valid	Valid	Valid
1025	Valid	Valid	Valid	Valid	Valid
1026	Valid	Valid	Valid	Valid	Valid
1027	Valid	Valid	Valid	Valid	Valid

CREATE DATE TABLE (DAX Table for Time Intelligence)

Step 1: Create a Date Table

Go to Home_Calculations → New Table

File Home Help Table tools Column tools

Clipboard

Get data workbook OneLake data hub SQL Server data Data

Transform Refresh data Queries

Manage relationships Relationships

New measure measure column Calculations

Quick New table

Manage roles View as Security

Sensitivity Sensitivity

Publish Share

id	address_state	application_type	emp_length	emp_title	grade	home_ownership	issue_date	last_credit_pull_date	last_payment_date	loan_status
331459	CA	INDIVIDUAL	7 years	Jabli Circuit	B	RENT	08 April 2021	16 May 2021	11 April 2021	Fully Paid
400097	CA	INDIVIDUAL	4 years	PMZ	B	RENT	09 May 2021	16 May 2021	12 May 2021	Fully Paid
401782	CA	INDIVIDUAL	2 years	Tri-City Technologies	B	RENT	09 July 2021	16 May 2021	12 July 2021	Fully Paid
416011	CA	INDIVIDUAL	3 years	Placer County Water Agency	B	RENT	09 June 2021	16 May 2021	12 July 2021	Fully Paid
417906	CA	INDIVIDUAL	5 years	Merced City School District	B	RENT	10 May 2021	16 May 2021	12 March 2021	Fully Paid
433888	CA	INDIVIDUAL	6 years	LA County Probation Dept.	B	RENT	09 August 2021	16 May 2021	11 October 2021	Fully Paid
441457	CA	INDIVIDUAL	2 years	International Checkout	B	RENT	09 September 2021	16 May 2021	12 May 2021	Fully Paid
449337	CA	INDIVIDUAL	9 years	Socket Mobile, Inc.	B	RENT	09 October 2021	16 May 2021	10 October 2021	Fully Paid
452391	CA	INDIVIDUAL	4 years	Siemens	B	RENT	09 November 2021	16 May 2021	11 July 2021	Fully Paid
453894	CA	INDIVIDUAL	4 years	Sprint	B	RENT	09 November 2021	16 May 2021	12 September 2021	Fully Paid

Data

Search

bank_loan_data

address_state

annual_income

application_type

dti

emp_length

emp_title

grade

File Home Help Table tools

Name Date Table

Structure

Mark as date table Calendars

Manage relationships Relationships

New measure measure column Calculations

Quick New table

1 Date Table = CALENDAR(MIN(bank_loan_data[issue_date]), MAX(bank_loan_data[issue_date]))

Date
13-11-2021 00:00:00
14-11-2021 00:00:00
15-11-2021 00:00:00
16-11-2021 00:00:00
17-11-2021 00:00:00
18-11-2021 00:00:00
19-11-2021 00:00:00
20-11-2021 00:00:00
21-11-2021 00:00:00
22-11-2021 00:00:00
23-11-2021 00:00:00
24-11-2021 00:00:00
25-11-2021 00:00:00
26-11-2021 00:00:00
27-11-2021 00:00:00
28-11-2021 00:00:00
29-11-2021 00:00:00
30-11-2021 00:00:00

Data

Search

bank_loan_data

address_state

annual_income

application_type

dti

emp_length

emp_title

grade

home_ownership

id

installment

int_rate

issue_date

last_credit_pull_date

File Home Help Table tools Column tools

Name Month

Data type Text

Format Text

Summarization Don't summarize

Data category Uncategorized

Sort by column Sort

Data groups Groups

Manage relationships Relationships

New column Calculations

1 Month = FORMAT('Date Table'[Date], "mmmm")

Date	Month
01-01-2021 00:00:00	January
02-01-2021 00:00:00	January
03-01-2021 00:00:00	January
04-01-2021 00:00:00	January
05-01-2021 00:00:00	January
06-01-2021 00:00:00	January
07-01-2021 00:00:00	January
08-01-2021 00:00:00	January
09-01-2021 00:00:00	January
10-01-2021 00:00:00	January
11-01-2021 00:00:00	January
12-01-2021 00:00:00	January
13-01-2021 00:00:00	January
14-01-2021 00:00:00	January
15-01-2021 00:00:00	January
16-01-2021 00:00:00	January
17-01-2021 00:00:00	January
18-01-2021 00:00:00	January
19-01-2021 00:00:00	January
20-01-2021 00:00:00	January
21-01-2021 00:00:00	January
22-01-2021 00:00:00	January
23-01-2021 00:00:00	January
24-01-2021 00:00:00	January

Data

Search

bank_loan_data

Date Table

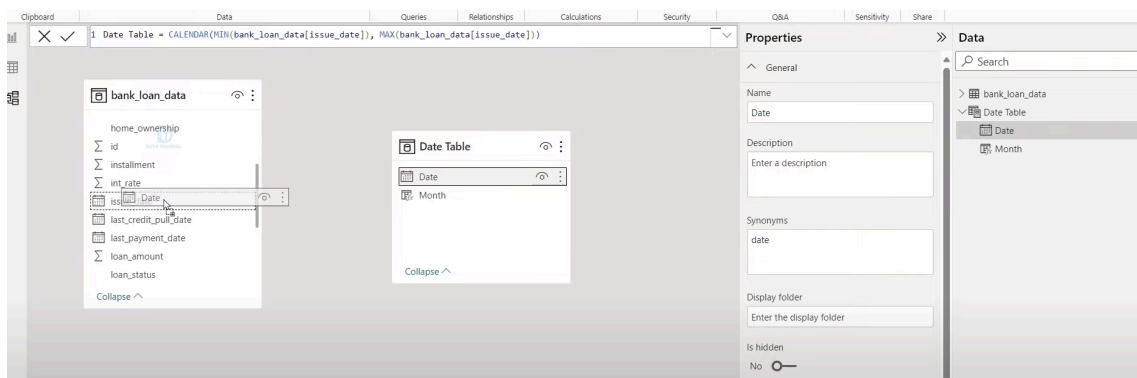
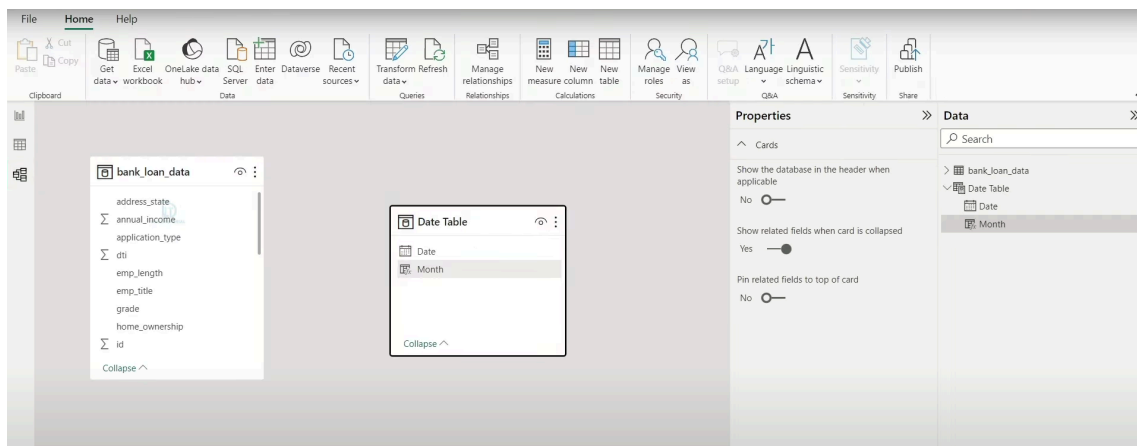
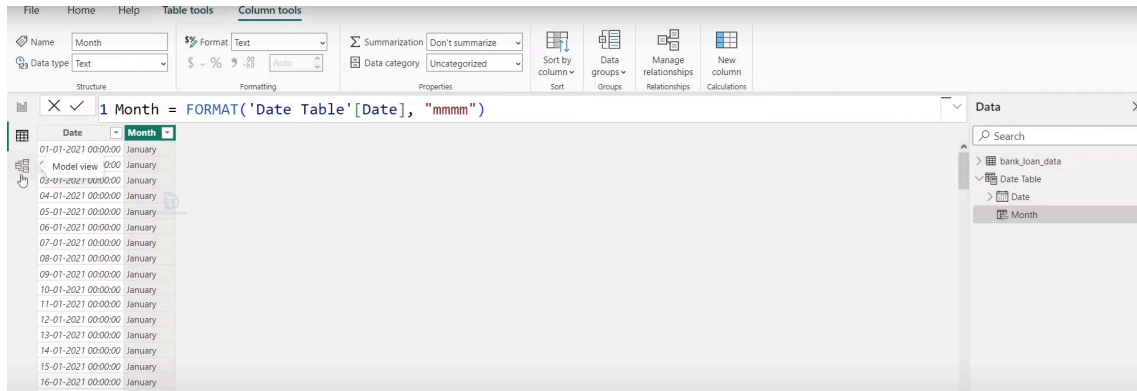
Date

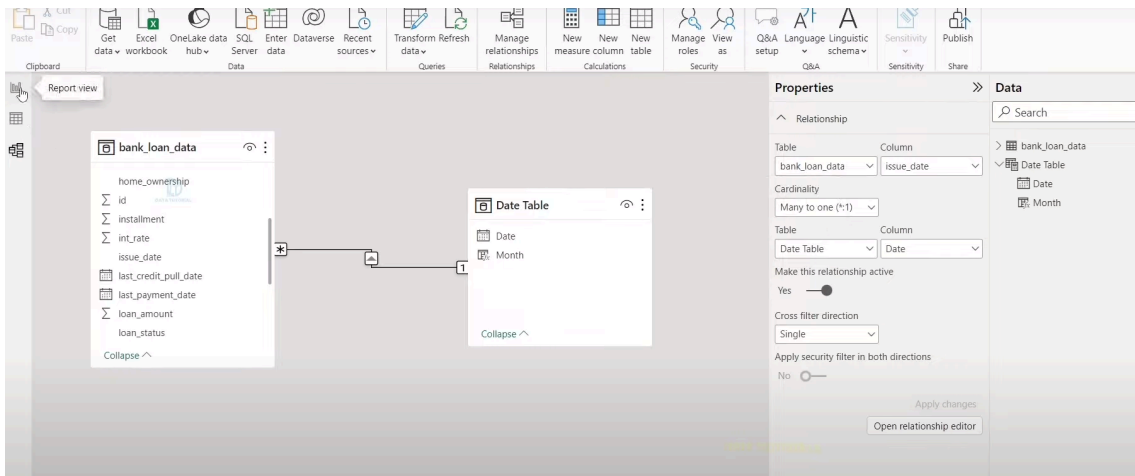
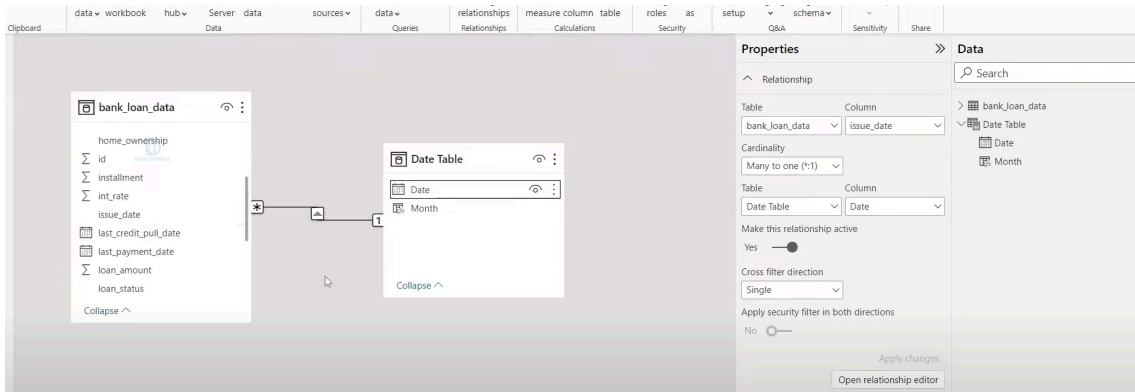
Month

Step 2: Establish Relationship (DATA MODELING)

- Set relationships:
 - Date Table (1) → Loan Data (many)**
 - Use appropriate **keys** (e.g., Loan_ID).

- Ensure relationship cardinality is **One-to-Many** and **Cross filter direction: Single**.
- In **Model view**, connect:
 - 'Date'[Date] → 'bank_loan_data'[issue_date]





DASHBOARD CREATION

Dashboard 1: SUMMARY (KPI Dashboard)

KPIs and DAX Calculations

PROBLEM STATEMENT



DASHBOARD 1: SUMMARY

Key Performance Indicators (KPIs) Requirements:

- 1. Total Loan Applications:** We need to calculate the total number of loan applications received during a specified period. Additionally, it is essential to monitor the Month-to-Date (MTD) Loan Applications and track changes Month-over-Month (MoM).
- 2. Total Funded Amount:** Understanding the total amount of funds disbursed as loans is crucial. We also want to keep an eye on the MTD Total Funded Amount and analyse the Month-over-Month (MoM) changes in this metric.
- 3. Total Amount Received:** Tracking the total amount received from borrowers is essential for assessing the bank's cash flow and loan repayment. We should analyse the Month-to-Date (MTD) Total Amount Received and observe the Month-over-Month (MoM) changes.
- 4. Average Interest Rate:** Calculating the average interest rate across all loans, MTD, and monitoring the Month-over-Month (MoM) variations in interest rates will provide insights into our lending portfolio's overall cost.
- 5. Average Debt-to-Income Ratio (DTI):** Evaluating the average DTI for our borrowers helps us gauge their financial health. We need to compute the average DTI for all loans, MTD, and track Month-over-Month (MoM) fluctuations.

Total Loan Applications

Base Measure (DAX)

```
Total Loan Applications = COUNT(bank_loan_data[id])
```

MTD Measure

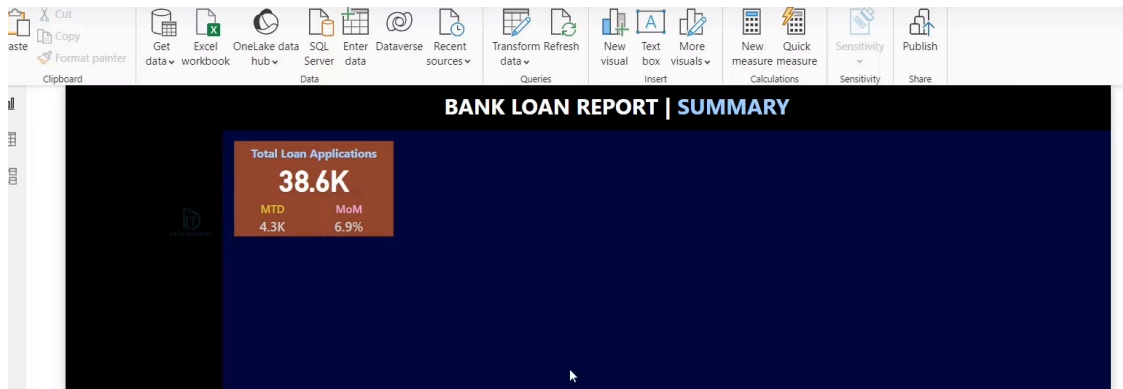
```
MTD Loan Applications = CALCULATE([Total Loan Applications], DATESMTD('Date Table'[Date]))
```

PMTD Measure

```
PMTD Loan Applications = CALCULATE([Total Loan Applications], DATESMTD(DATEADD('Date Table'[Date], -1, MONTH)))
```

MoM Change Measure

MoM Loan Applications =
 ([MTD Loan Applications] - [PMTD Loan Applications]) / [PMTD Loan Applications]



Total Funded Amount

Base Measure (DAX)

Total Funded Amount = SUM(bank_loan_data[loan_amount])

MTD Measure

MTD Funded Amount = CALCULATE([Total Funded Amount], DATESMTD('Date Table'[Date]))

PMTD Measure

PMTD Funded Amount = CALCULATE([Total Funded Amount], DATESMTD(DATEADD('Date Table'[Date], -1, MONTH)))

MoM Change Measure

MoM Funded Amount =
([MTD Funded Amount] - [PMTD Funded Amount]) / [PMTD Funded Amount]

Total Amount Received

Base Measure (DAX)

Total Amount Received = SUM(bank_loan_data[total_payment])

MTD Measure

MTD Amount Received = CALCULATE([Total Amount Received], DATESMTD('Date Table'[Date]))

PMTD Measure

PMTD Amount Received = CALCULATE([Total Amount Received], DATESMTD(DATEADD('Date Table'[Date], -1, MONTH)))

MoM Change Measure

MoM Amount Received =
([MTD Amount Received] - [PMTD Amount Received]) / [PMTD Amount Received]

Average Interest Rate

Base Measure (DAX)

Average Interest Rate = AVERAGE(bank_loan_data[int_rate])

MTD Measure

MTD Average Interest Rate = CALCULATE([Average Interest Rate], DATESMTD('Date Table'[Date]))

PMTD Measure

PMTD Average Interest Rate = CALCULATE([Average Interest Rate], DATESMTD(DATEADD('Date Table'[Date], -1, MONTH)))

MoM Change Measure

MoM Average Interest Rate =
([MTD Average Interest Rate] - [PMTD Average Interest Rate]) / [PMTD Average Interest Rate]

Average DTI

Base Measure (DAX)

Average DTI = AVERAGE(bank_loan_data[dti])

MTD Measure

MTD Average DTI = CALCULATE([Average DTI], DATESMTD('Date Table'[Date]))

PMTD Measure

PMTD Average DTI = CALCULATE([Average DTI], DATESMTD(DATEADD('Date Table'[Date], -1, MONTH)))

MoM Change Measure

MoM Average DTI =
([MTD Average DTI] - [PMTD Average DTI]) / [PMTD Average DTI]



Include Good Loan vs Bad Loan Metrics:

- Good Loan = Fully Paid / Current
- Bad Loan = Charged Off / Default

- ✓ Use **Card visuals** for KPIs
- ✓ Use **Grid Table** for Loan Status comparison

PROBLEM STATEMENT

DASHBOARD 1: SUMMARY



Good Loan v Bad Loan KPI's

Good Loan:

1. Good Loan Application Percentage
2. Good Loan Applications
3. Good Loan Funded Amount
4. Good Loan Total Received Amount

Bad Loan

1. Bad Loan Application Percentage
2. Bad Loan Applications
3. Bad Loan Funded Amount
4. Bad Loan Total Received Amount



Groups

×

Name *

loan_status (groups)

Field

loan_status

Group type

List

Ungrouped values

Charged Off

Current

Fully Paid

Groups and members

Group

Ungroup

☐ Include Other group ⓘ

OK

Cancel

Groups

Name *

loan_status (groups)

Field

loan_status

Group type

List

Ungrouped values

Charged Off

Current

Fully Paid

Groups and members

Group

Ungroup

☐ Include Other group ⓘ

OK

Cancel

Groups

Name *

loan_status (groups)

Field

loan_status

Group type

List

Ungrouped values

Groups and members

▸ Bad Loan

◦ Charged Off

▸ Good Loan

◦ Current

◦ Fully Paid

Group

Ungroup

☐ Include Other group ⓘ

Cancel

Groups ✕

Name *

Field

Group type

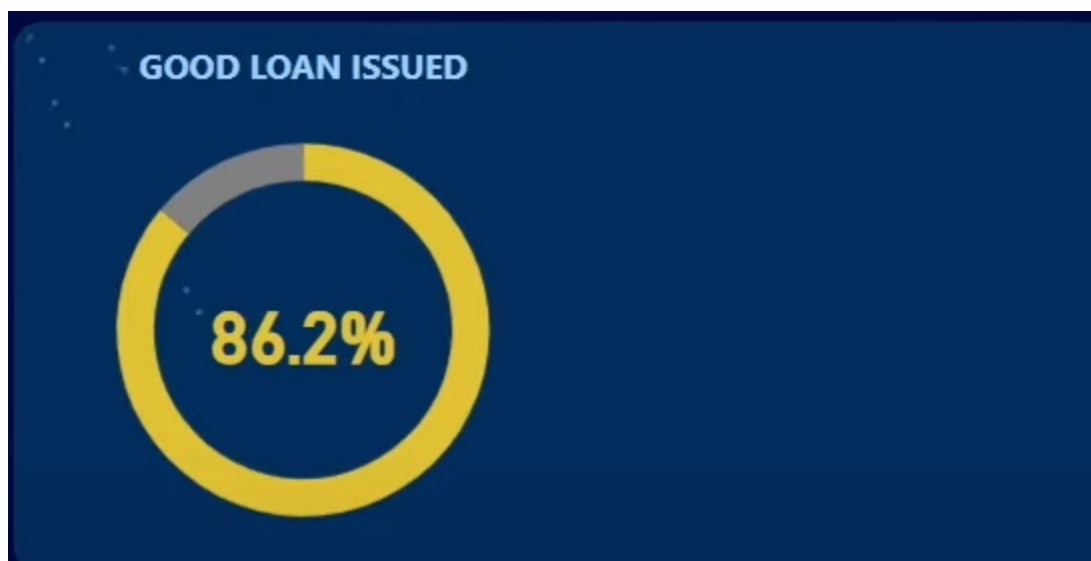
Ungrouped values

Groups and members

- ▶ Bad Loan
- ▶ Good Loan

☐ Include Other group ⓘ

```
1 Good Loan % = (CALCULATE([Total Loan Applications], bank_loan_data[Good v Bad Loan] = "Good Loan")) / [Total Loan Applications]
```



```

1 Good Loan Applications = CALCULATE([Total Loan Applications], bank_loan_data[Good v Bad
  Loan] = "Good Loan")

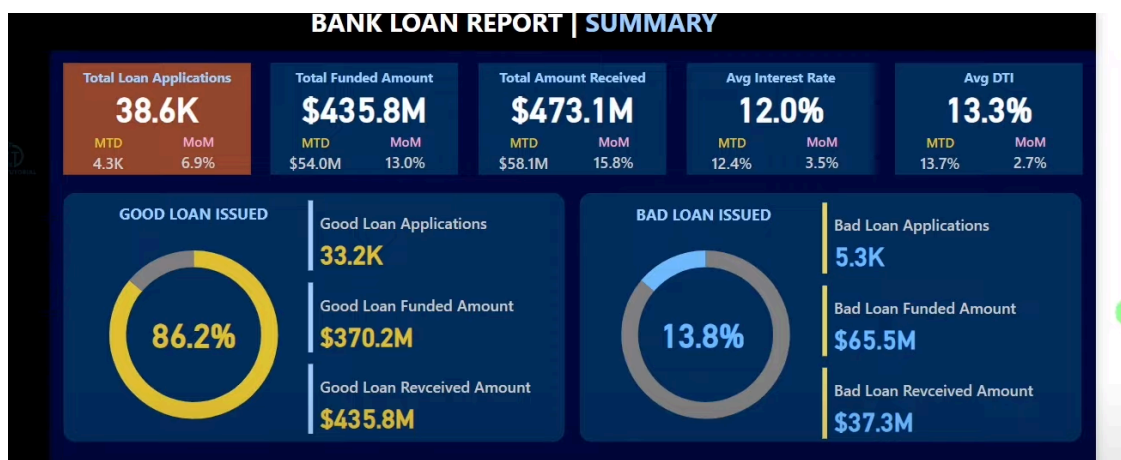
1 Good Loan Funded Amount = CALCULATE([Total Funded Amount], bank_loan_data[Good v Bad Loan]
  = "Good Loan")

1 Good Loan Revceived Amount = CALCULATE([Total Amount Received], bank_loan_data[Good v Bad
  Loan] = "Good Loan")

```

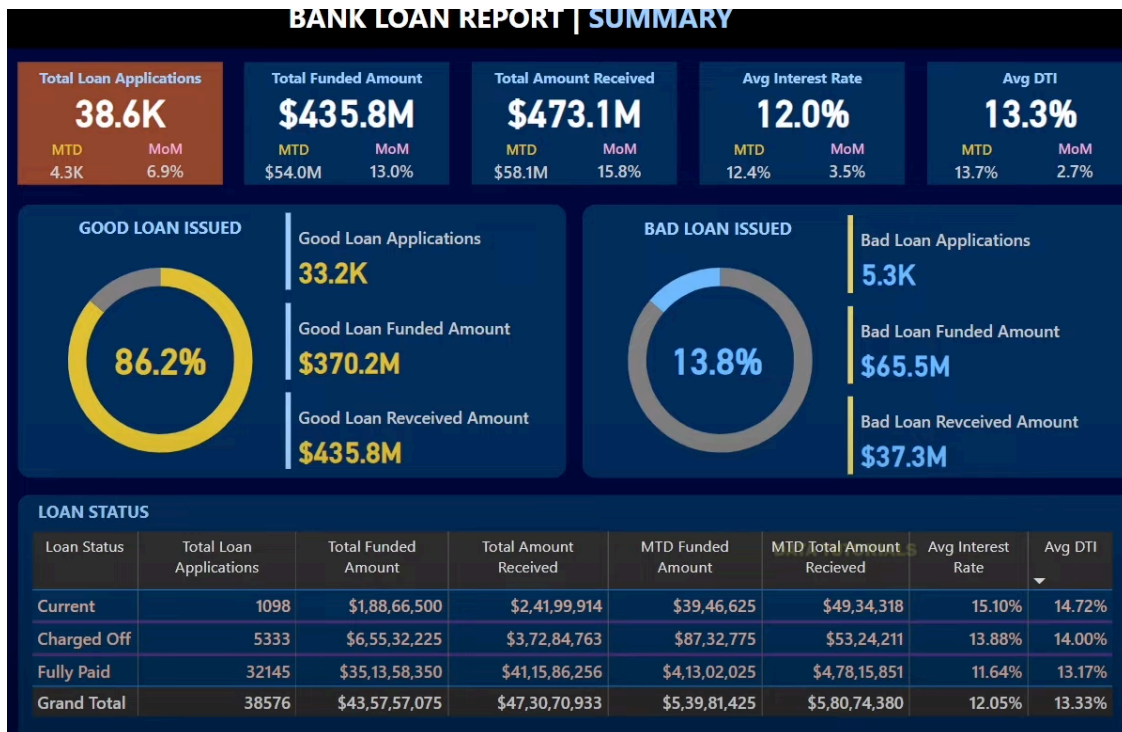


SAME STEPS FOR BAD LOAN ONLY CHANGE Good Loan——Bad Loan

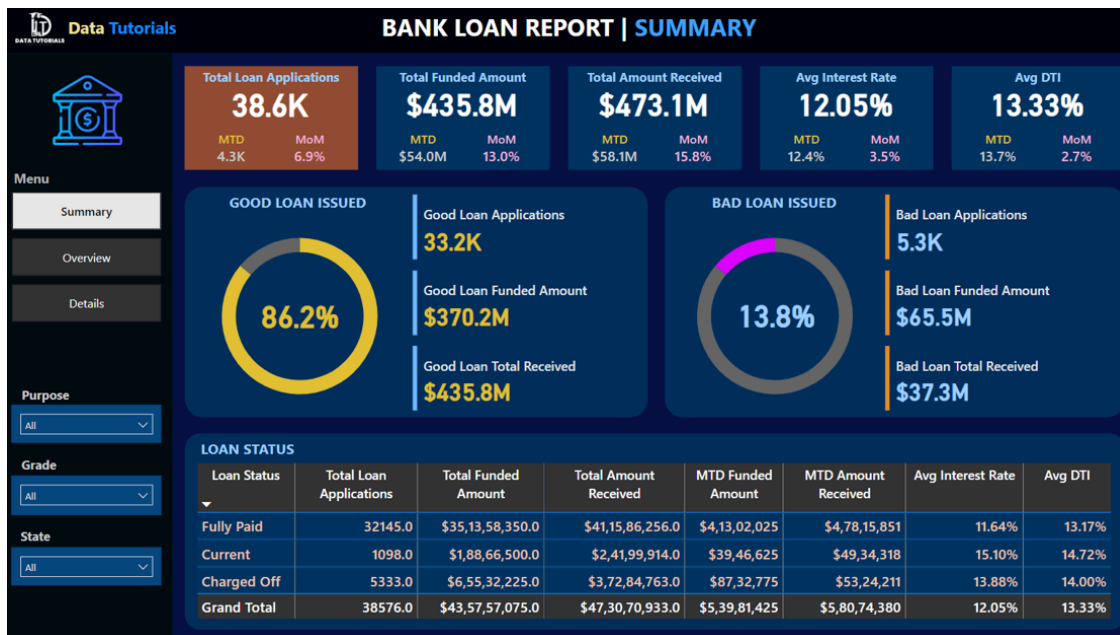
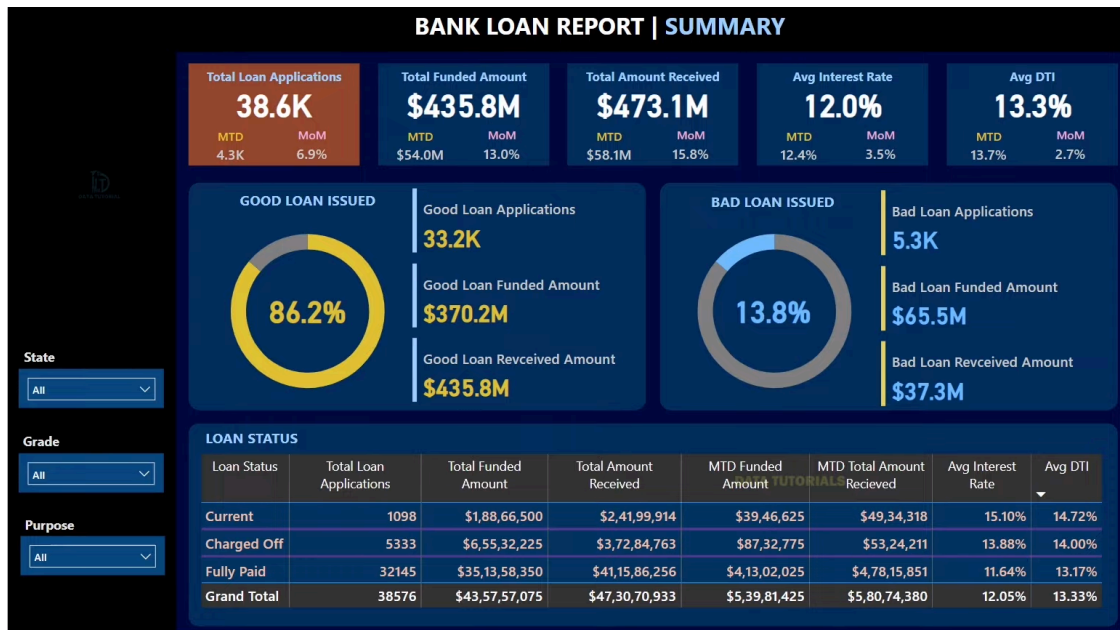


Loan Status Grid View

In order to gain a comprehensive overview of our lending operations and monitor the performance of loans, we aim to create a grid view report categorized by 'Loan Status.' By providing insights into metrics such as 'Total Loan Applications,' 'Total Funded Amount,' 'Total Amount Received,' 'Month-to-Date (MTD) Funded Amount,' 'MTD Amount Received,' 'Average Interest Rate,' and 'Average Debt-to-Income Ratio (DTI),' this grid view will empower us to make data-driven decisions and assess the health of our loan portfolio.



Adding Slicers



Dashboard 2: OVERVIEW (Exploratory Visuals)

PROBLEM STATEMENT



DASHBOARD 2: OVERVIEW

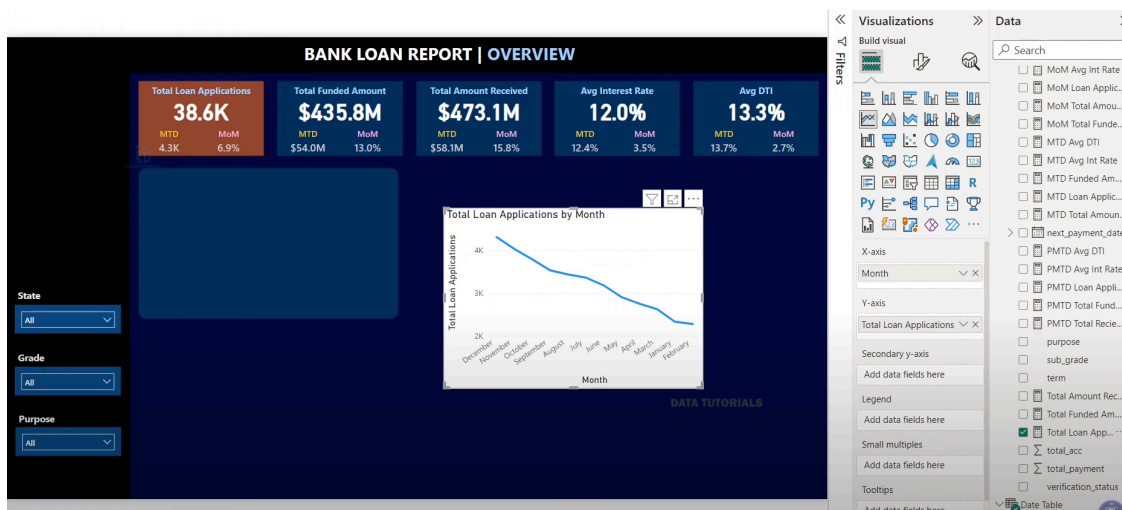
CHARTS

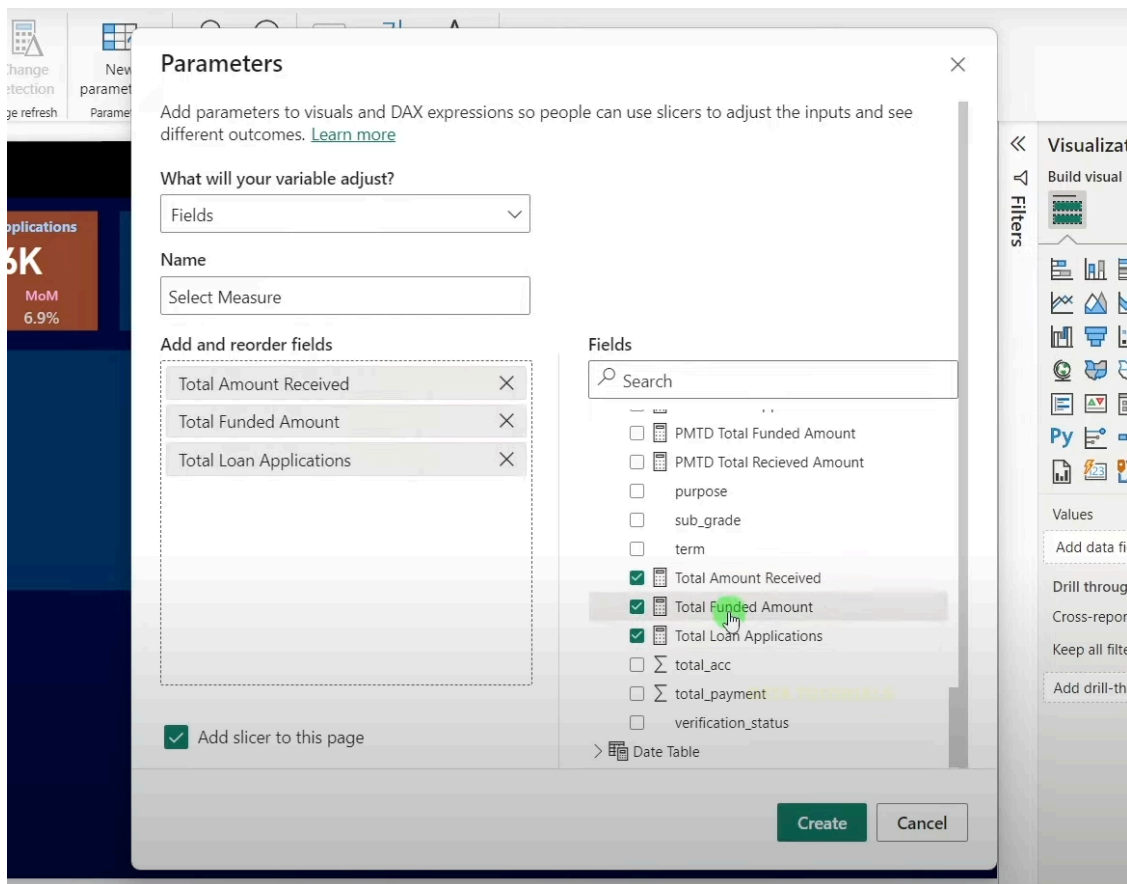
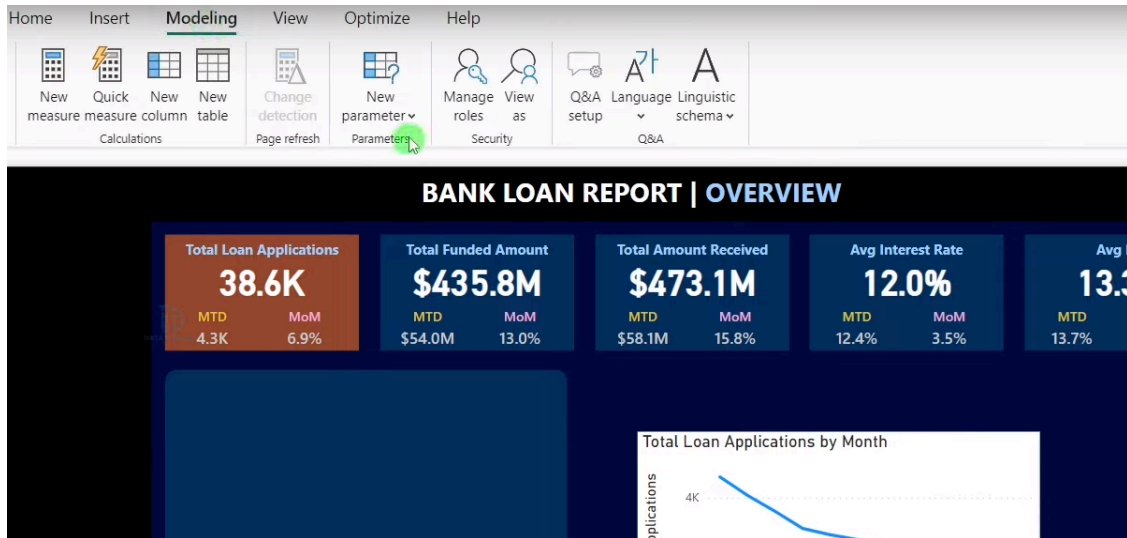
1. **Monthly Trends by Issue Date (Line Chart):** To identify seasonality and long-term trends in lending activities
2. **Regional Analysis by State (Filled Map):** To identify regions with significant lending activity and assess regional disparities
3. **Loan Term Analysis (Donut Chart):** To allow the client to understand the distribution of loans across various term lengths.
4. **Employee Length Analysis (Bar Chart):** How lending metrics are distributed among borrowers with different employment lengths, helping us assess the impact of employment history on loan applications.
5. **Loan Purpose Breakdown (Bar Chart):** Will provide a visual breakdown of loan metrics based on the stated purposes of loans, aiding in the understanding of the primary reasons borrowers seek financing.
6. **Home Ownership Analysis (Tree Map):** For a hierarchical view of how home ownership impacts loan applications and disbursements.

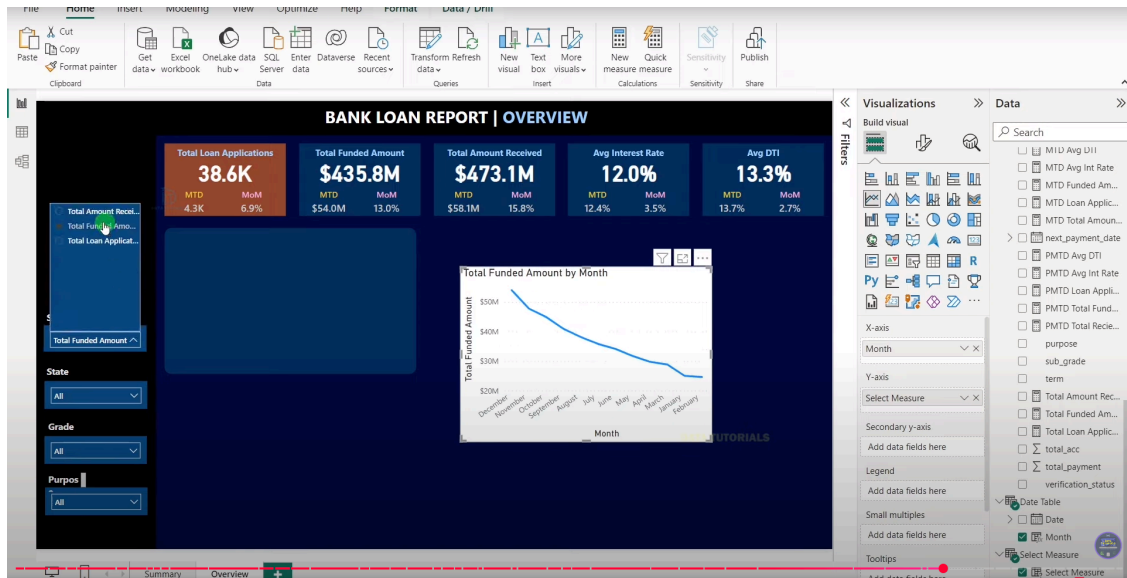
Metrics to be shown: 'Total Loan Applications,' 'Total Funded Amount,' and 'Total Amount Received'

Monthly Trends by Issue Date

Field Parameter



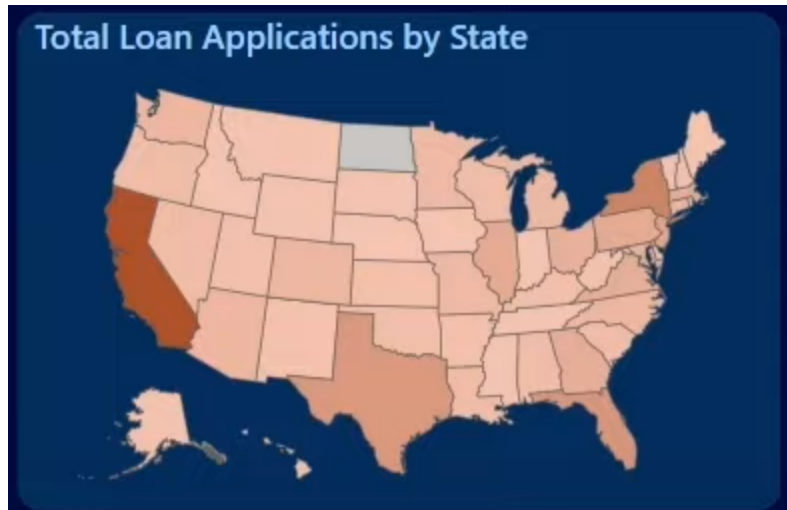




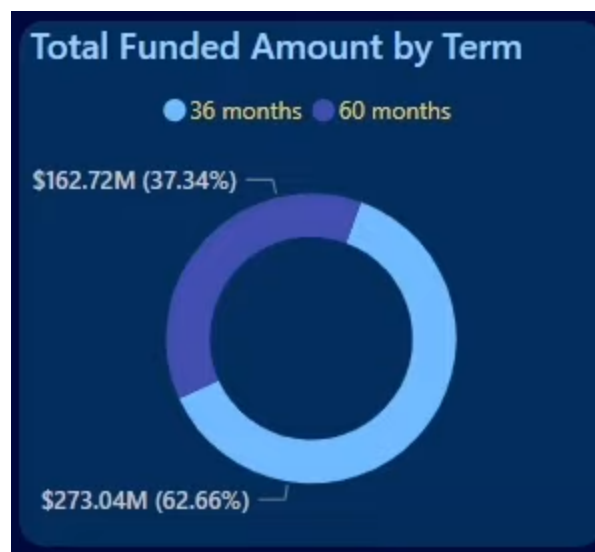
1 Month number = MONTH('Date Table'[Date])

Date	Month	Month number
04-01-2021 00:00:00	Jan	1
05-01-2021 00:00:00	Jan	1
06-01-2021 00:00:00	Jan	1
07-01-2021 00:00:00	Jan	1
08-01-2021 00:00:00	Jan	1
09-01-2021 00:00:00	Jan	1
10-01-2021 00:00:00	Jan	1
11-01-2021 00:00:00	Jan	1
12-01-2021 00:00:00	Jan	1
13-01-2021 00:00:00	Jan	1
14-01-2021 00:00:00	Jan	1
15-01-2021 00:00:00	Jan	1
16-01-2021 00:00:00	Jan	1
17-01-2021 00:00:00	Jan	1
18-01-2021 00:00:00	Jan	1
19-01-2021 00:00:00	Jan	1
20-01-2021 00:00:00	Jan	1
21-01-2021 00:00:00	Jan	1
22-01-2021 00:00:00	Jan	1
23-01-2021 00:00:00	Jan	1
24-01-2021 00:00:00	Jan	1
25-01-2021 00:00:00	Jan	1
26-01-2021 00:00:00	Jan	1
27-01-2021 00:00:00	Jan	1
28-01-2021 00:00:00	Jan	1
29-01-2021 00:00:00	Jan	1

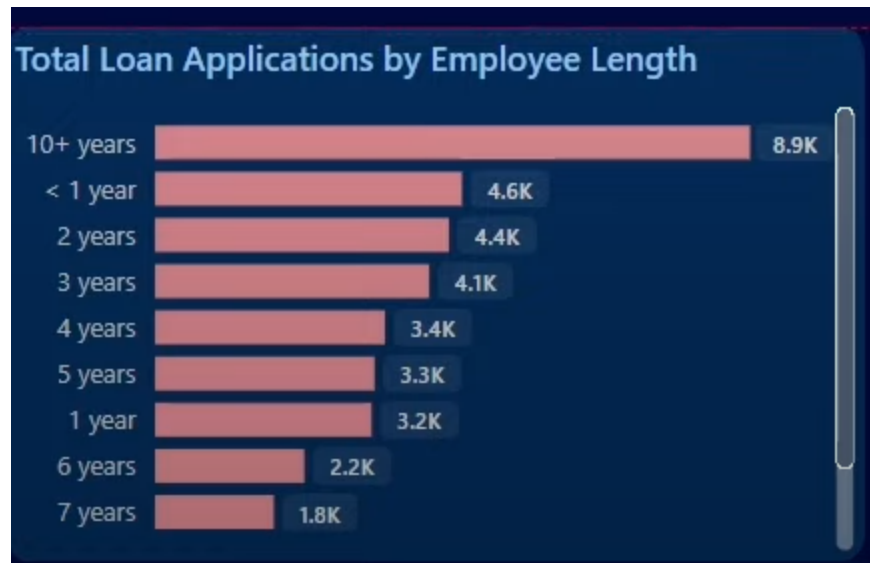
Regional Analysis by State



Loan Term Analysis



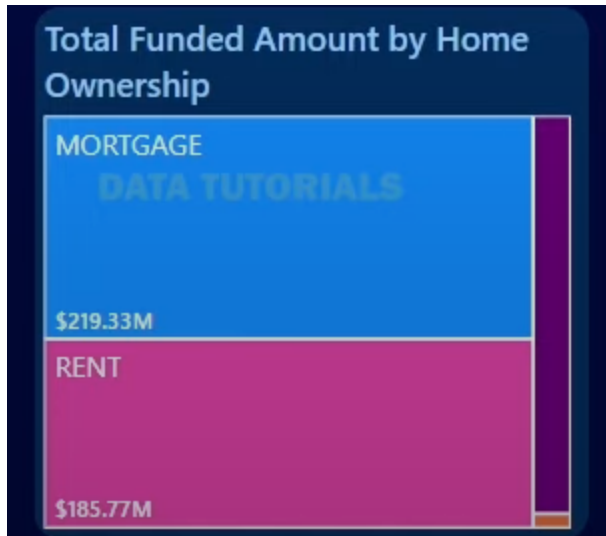
Employee Length Analysis



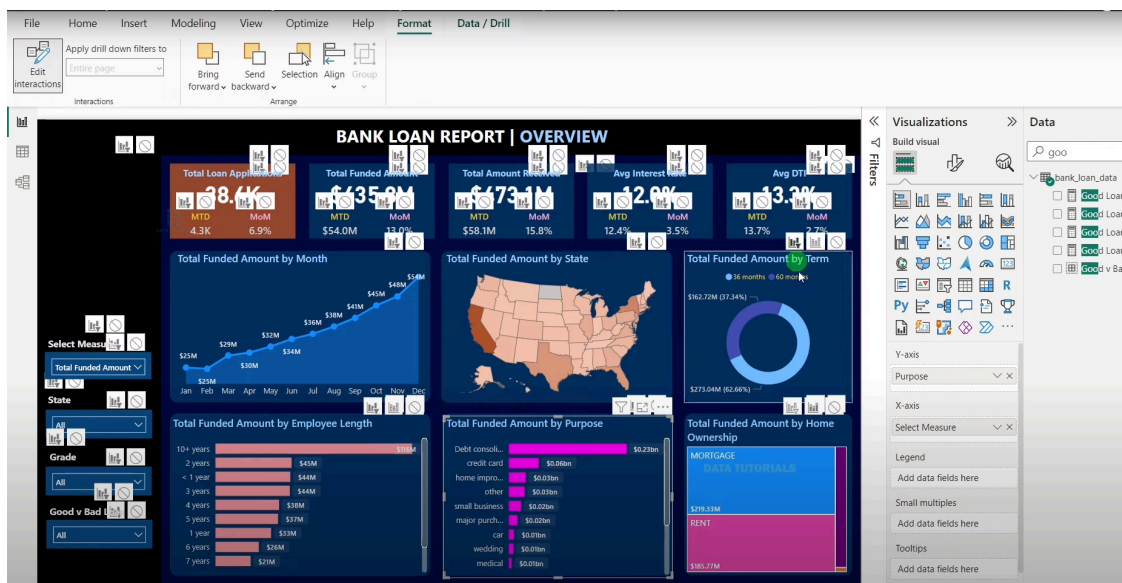
Loan Purpose Breakdown



Home Ownership Analysis



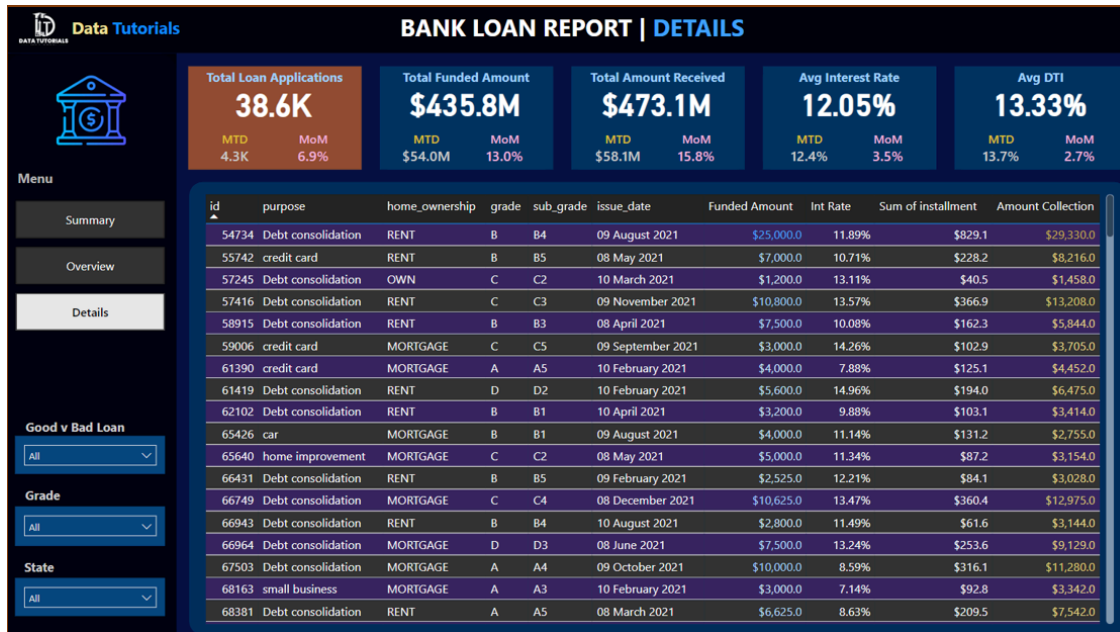
Filter Interactions





Dashboard 3: DETAILS (Data Grid Dashboard)

- Create a detailed table view with:
 - Loan ID, Borrower Info, Loan Amount, DTI, Status, Issue Date, Term, etc.
- Add **Slicers**:
 - Loan Status, Purpose, DTI range, State.
- Provide a full 360° view of loan performance.



Importance of Data Analysis in Banking

1. Risk Assessment

Evaluating borrower creditworthiness to minimize defaults and ensure responsible lending practices.

2. Decision-Making

Empowering banks to make data-driven decisions on loan approvals, interest rates, and borrower segmentation.

3. Portfolio Management

Monitoring loan portfolio performance to identify underperforming or high-risk loans and maintain portfolio health.

4. Fraud Detection

Detecting unusual application patterns, inconsistencies, or suspicious behavior to prevent fraudulent loan activities.

5. Regulatory Compliance

Ensuring adherence to financial regulations and reporting standards required by governing authorities.

Conclusion

This project demonstrates the importance of **data analysis in banking**, showcasing how **Power BI visualizations** and **SQL-driven insights** enable better understanding of **loan performance**, enhance **risk control**, and support **strategic decision-making**.

Interview Explanation Script — “Bank Loan Analysis Project”

1 Introduction & Tools

In this project, I analyzed a large bank-loan dataset to understand lending trends, borrower behavior, and overall loan performance.

The project was built using **MS SQL Server** for data extraction and cleaning, and **Power BI** for visualization and dashboard creation.

2 Objective

The main objective was to identify factors influencing loan approval, repayment performance, and risk assessment.

I also wanted to visualize metrics such as total loan applications, funded amount, repayment trends, and borrower credit health through interactive dashboards.

3 Data Understanding & Cleaning

The data contained fields like Loan ID, Issue Date, Loan Amount, Interest Rate, DTI, Employment Length, Home Ownership, and Loan Status.

After connecting Power BI to SQL Server, I used **Power Query Editor** to check **column quality, distribution, and missing values**.

I removed duplicates, handled nulls, corrected data types, and created a clean dataset for analysis.

4 Data Modeling

I created a Date Table using DAX and linked it to the Issue Date field to enable time intelligence functions such as MTD, PMTD, and MoM analysis.

Relationships were built in the model view — one-to-many between the Date table and the loan data table.

5 KPI & DAX Calculations

I created several Key Performance Indicators (KPIs):

- Total Loan Applications
- Total Funded Amount
- Total Amount Received
- Average Interest Rate
- Average DTI

For each of these, I calculated **Month-to-Date (MTD)**, **Previous Month-to-Date (PMTD)**, and **Month-over-Month (MoM)** growth using **CALCULATE** and **DATESMTD** functions in DAX.

Example:

```
MTD Loan Applications = CALCULATE([Total Loan Applications], DATESMTD('Date Table'[Date]))
```

6 Dashboard Creation

I designed three dashboards in Power BI:

✓ **Summary Dashboard** – showed KPIs like total loans, funded amounts, repayments, and interest rate trends, along with good vs bad loan performance.

✔ **Overview Dashboard** – visualized insights using:

Line chart for monthly trends
Map for regional loan activity
Donut chart for loan term distribution
Bar charts for employment length and loan purpose
Tree map for home ownership analysis

✔ **Details Dashboard** – provided a grid view of all loans with slicers for filtering by status, purpose, or DTI range.

7 Insights & Outcomes

From the analysis, I observed clear patterns such as:

- Borrowers with stable employment and lower DTI had higher repayment rates.
- Certain states had higher default ratios, useful for regional risk assessment.
- Most loans were taken for debt consolidation and home improvement purposes.

8 Business Impact / Conclusion

Overall, this project demonstrated how data analytics helps banks improve risk assessment, decision-making, fraud detection, and portfolio management.

By combining SQL and Power BI, I was able to transform raw loan data into clear, actionable insights for management.

💡 **Optional 20-second Summary (if interviewer asks for a quick version)**

"This project analyzed bank loan data using SQL and Power BI.

I cleaned and modeled the data, created DAX measures for key KPIs like loan applications, funded amount, and interest rate, and built interactive dashboards to track MTD and MoM performance.

The insights helped assess borrower risk, identify regional trends, and improve lending decisions."

Challenges Faced During the Bank Loan Analysis Project

While developing the **Bank Loan Analysis** project using **Python (Jupyter Notebook)** and **SQL**, I encountered several real-world challenges that helped me build both technical and analytical skills.

1. Data Cleaning and Preprocessing

Challenge:

The raw dataset contained missing values, inconsistent data formats, and duplicate entries — especially in fields like `employee_length`, `annual_income`, and `loan_status`.

How I Handled It:

- Used **Pandas** in Python to identify and handle missing data (`.isnull()`, `.fillna()`, `.dropna()`).
 - Converted **data types** (e.g., date columns to `datetime`, numeric columns to `float/int`).
 - Standardized categorical fields (like "Fully Paid", "fully paid", "Paid" → "Fully Paid").
 - Removed **duplicate records** to ensure accuracy in analysis.
-

2. Data Integration Between SQL and Python

Challenge:

Managing and verifying the same dataset in both **SQL (for storage & validation)** and **Python (for analysis & visualization)** required careful synchronization.

How I Handled It:

- Created the `bank_loan_data` table in SQL Server.
 - Used Python's `pyodbc` library to connect Jupyter Notebook with SQL Server.
 - Cross-checked record counts and column types after every import to ensure no data loss or mismatch.
-

3. Understanding Financial Terminologies

Challenge:

Initially, it was difficult to understand banking terms like **DTI (Debt-to-Income Ratio)**, **Sub-Grade**, **Verification Status**, and how they impact risk assessment.

How I Handled It:

- Researched each field using banking and finance documentation.
 - Created a **Terminology Dictionary** section in my project (explaining each field's purpose and use for banks).
 - This improved my domain knowledge and helped me interpret insights correctly.
-

4. Feature Engineering and Insights Generation

Challenge:

Identifying which features were most relevant for analysis and KPI creation (e.g., **good loan ratio**, **average interest rate**, **default percentage**) was tricky.

How I Handled It:

- Created calculated fields using **Pandas** and **NumPy**.
 - Segmented data by `loan_status` to generate metrics like:
 - Total good loans vs bad loans
 - Average interest rate per grade
 - Month-to-date and month-over-month loan trends
-



5. Visualization and Interpretation

Challenge:

Selecting appropriate charts to visualize loan trends, default patterns, and income distributions.

How I Handled It:

- Used **Matplotlib** and **Seaborn** to create clear and interactive visuals.
 - Focused on storytelling — e.g., visualizing how **annual income vs DTI** affects loan approval.
-



6. Time-Based KPIs (MTD, PMTD, MoM)

Challenge:

Calculating **Month-to-Date**, **Previous Month-to-Date**, and **Month-over-Month** metrics required accurate date handling.

How I Handled It:

- Used **Python datetime functions** and **Pandas** `to_datetime()` for consistency.
 - Validated calculations using SQL queries for cross-checking accuracy.
-



7. Project Structuring and Documentation

Challenge:

Combining SQL workflow, Python code, and explanation into a single cohesive project was time-consuming.

How I Handled It:

- Documented each phase in **Markdown** (for Notion and Jupyter Notebook).
- Divided project into clear **phases**:

1 Data Understanding

- 2 Data Cleaning
 - 3 SQL Handling
 - 4 KPI Creation
 - 5 Visualization & Insights
-

8. Realization / Key Learning

This project taught me:

- How to handle end-to-end data pipelines (SQL → Python → Visualization).
- Importance of data validation and domain understanding.
- How to derive meaningful insights that support **banking decisions** like **risk assessment** and **loan approval strategy**.

| THANKS |